

Applied Intelligence

VoxelDiffusionCut: Non-destructive Internal-part Extraction via Iterative Cutting and Structure Estimation

--Manuscript Draft--

Manuscript Number:		
Full Title:	VoxelDiffusionCut: Non-destructive Internal-part Extraction via Iterative Cutting and Structure Estimation	
Article Type:	Original Submission	
Keywords:	Autonomous Dismantling; Waste Management; Structure Estimation	
Corresponding Author:	Tomoya Yamanokuchi, Ph.D. Nara Institute of Science and Technology: Nara Sentan Kagaku Gijutsu Daigakuin Daigaku Ikoma, Nara JAPAN	
Corresponding Author Secondary Information:		
Corresponding Author's Institution:	Nara Institute of Science and Technology: Nara Sentan Kagaku Gijutsu Daigakuin Daigaku	
Corresponding Author's Secondary Institution:		
First Author:	Takumi Hachimine, M.S.	
First Author Secondary Information:		
Order of Authors:	Takumi Hachimine, M.S.	
	Yuhwan Kwon, Ph.D.	
	Cheng-Yu Kuo, Ph.D.	
	Tomoya Yamanokuchi, Ph.D.	
	Takamitsu Matsubara, Ph.D.	
Order of Authors Secondary Information:		
Funding Information:	New Energy and Industrial Technology Development Organization (JPNP23002)	Prof. Takamitsu Matsubara
Abstract:	Non-destructive extraction of the target internal part, such as batteries and motors, by cutting surrounding structures is crucial at recycling and disposal sites. However, the diversity of products and the lack of information on disassembly procedures make it challenging to decide where to cut. This study explores a method for non-destructive extraction of a target internal part that iteratively estimates the internal structure from observed cutting surfaces and formulates cutting plans based on the estimation results. A key requirement is to estimate the probability of the target part's presence from partial observations. However, learning conditional generative models for this task is challenging: The high dimensionality of 3D shape representations makes learning difficult, and conventional models (e.g., conditional variational autoencoders) often fail to capture multi-modal predictive uncertainty due to mode collapse, resulting in overconfident predictions. To address these issues, we propose VoxelDiffusionCut, which iteratively estimates the internal structure represented as voxels using a diffusion model and plans cuts for non-destructive extraction of the target internal part based on the estimation results. Voxel representation allows the model to predict only attributes at fixed grid positions, i.e., types of constituent parts, making learning more tractable. The diffusion model completes the voxel representation conditioned on observed cutting surfaces, capturing uncertainty in unobserved regions to avoid erroneous cuts. Experimental results in simulation suggest that the proposed method can estimate internal structures from observed cutting surfaces and enable non-destructive extraction of the target internal part by leveraging the estimated uncertainty.	

VoxelDiffusionCut: Non-destructive Internal-part Extraction via Iterative Cutting and Structure Estimation

Takumi Hachimine¹, Yuhwan Kwon¹, Cheng-Yu Kuo¹,
Tomoya Yamanokuchi^{1*}, Takamitsu Matsubara¹

¹Division of Information Science, Graduate School of Science and
Technology, Nara Institute of Science and Technology, 8916-5
Takayama, Ikoma, 630-0192, Nara, Japan.

*Corresponding author(s). E-mail(s): yamanokuchi.tomoya@naist.ac.jp;
Contributing authors: hachimine.takumi.hs8@is.naist.jp;
y-kwon@kansai-u.ac.jp; chengyu.kuo@naist.ac.jp; takam-m@is.naist.jp;

Abstract

Non-destructive extraction of the target internal part, such as batteries and motors, by cutting surrounding structures is crucial at recycling and disposal sites. However, the diversity of products and the lack of information on disassembly procedures make it challenging to decide where to cut. This study explores a method for non-destructive extraction of a target internal part that iteratively estimates the internal structure from observed cutting surfaces and formulates cutting plans based on the estimation results. A key requirement is to estimate the probability of the target part's presence from partial observations. However, learning conditional generative models for this task is challenging: The high dimensionality of 3D shape representations makes learning difficult, and conventional models (e.g., conditional variational autoencoders) often fail to capture multi-modal predictive uncertainty due to mode collapse, resulting in overconfident predictions. To address these issues, we propose VoxelDiffusionCut, which iteratively estimates the internal structure represented as voxels using a diffusion model and plans cuts for non-destructive extraction of the target internal part based on the estimation results. Voxel representation allows the model to predict only attributes at fixed grid positions, i.e., types of constituent parts, making learning more tractable. The diffusion model completes the voxel representation conditioned on observed cutting surfaces, capturing uncertainty in unobserved regions to avoid erroneous cuts. Experimental results in simulation

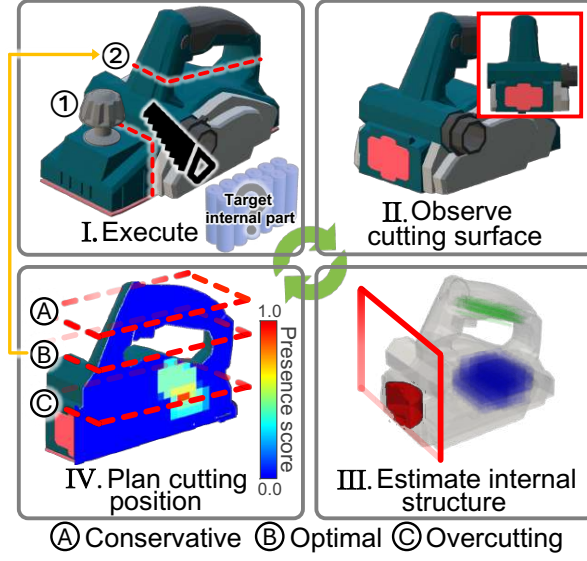


Fig. 1. Overview of the process for non-destructive extraction of the target internal part. The process iteratively estimates the internal structure from the observed cutting surfaces and plans cutting positions to extract the target part without damage.

suggest that the proposed method can estimate internal structures from observed cutting surfaces and enable non-destructive extraction of the target internal part by leveraging the estimated uncertainty.

Keywords: Autonomous Dismantling, Waste Management, Structure Estimation

1 Introduction

Extracting internal parts within products is an essential task at recycling and disposal sites [1, 2]. These parts, such as batteries, motors, and electronic components, must be extracted without damage to ensure safety and their functionality. If product information is available, internal parts can be extracted through disassembly [3–6]. However, such information is rarely disclosed due to concerns about intellectual property, and disassembly is often infeasible due to adhesives or degraded joints. Even when products share the same external shape, their internal structures may vary due to minor modifications introduced in different production years or specifications, making part extraction difficult.

Mechanical separation, such as *cutting*, is a practical method applicable even for products with adhesives or degradation [7]; however, deciding where to cut remains challenging due to the unknown internal structure. Therefore, we explore a method for non-destructive extraction of the target internal part from a product with an unknown internal structure, based on cutting surfaces observed during the cutting process (Fig. 1). As a first step toward such automation, this study focuses on the

1 extraction of a target internal part from products with identical external shapes but
2 different internal structures. To focus on the fundamental challenges inherent in the
3 problem, we adopt an idealized setting without modeling observation noise or cutting
4 errors.¹

5 A key difficulty in non-destructive extraction of the target internal part lies in
6 estimating the entire internal structure from partial observations on cutting surfaces.
7 Therefore, we consider an iterative approach: (a) estimating the internal structure
8 based on observed cutting surfaces, and (b) planning cutting positions that maxi-
9 mize removable volume while avoiding damage to the target internal part. To achieve
10 non-destructive extraction, it is necessary to estimate the probability of the target
11 part’s presence from partial observations, which entails estimating the uncertainty of
12 the internal structure in the unobserved regions. This estimation can be viewed as
13 a conditional generative modeling problem, where internal structures are generated
14 conditioned on partial observations. However, learning such models poses two chal-
15 lenges: (1) Directly learning internal structures as 3D shapes (e.g., point clouds or
16 meshes) is difficult due to the high dimensionality of these representations and their
17 unordered data structures [8–10], which makes it hard to reproduce internal structures
18 sufficiently and can lead to erroneous cuts. (2) Conventional generative models, such
19 as Conditional Variational Autoencoders (CVAEs) [11] and Conditional Generative
20 Adversarial Networks (CGANs) [12] often suffer from failure to capture multi-modal
21 predictive uncertainty due to mode collapse, resulting in overconfident predictions.

22 To address these issues, we view internal structure estimation as 3D shape comple-
23 tion in *voxel space*, where the observed cutting surfaces provide partial observations
24 of the voxelized shape. By representing 3D shapes as voxels and encoding constituent
25 parts as voxel attributes, the model only needs to predict these attributes at fixed
26 grid positions. This regular grid structure can make learning more tractable compared
27 to directly handling irregular 3D geometry. We formulate this voxel completion prob-
28 lem using recent generative models, such as diffusion models [13–16], conditioned on
29 the observed cutting surfaces. Diffusion models flexibly capture multi-modal predic-
30 tive uncertainty in unobserved regions through iterative denoising-based generation,
31 thereby mitigating mode collapse and alleviating overconfident predictions, which
32 helps reduce the risk of incorrect cutting plans.

33 With the above in mind, we propose VoxelDiffusionCut, a framework for non-
34 destructive extraction of a target internal part from a product with an unknown
35 internal structure, using observed cutting surfaces. VoxelDiffusionCut consists of iter-
36 ative internal structure estimation in a voxel representation using a diffusion model
37 and cutting planning. The estimation results are incorporated in the cutting plan to
38 maximize removable volume while avoiding damage to the target internal part.

39 We evaluated the proposed method in a simulator using simple-shaped models and
40 complex-shaped models that emulate real products. The experimental results suggest
41 that our method can estimate internal structures from observed cutting surfaces while
42 capturing uncertainty in unobserved regions. Moreover, cutting plans based on these
43 estimations enable non-destructive extraction of the target internal part.

44 The key contributions of this paper are as follows:

45 ¹Practical extensions are discussed in §6.

- We formulate a new problem setting for product dismantling, in which a target internal part is extracted by cutting a product with an unknown internal structure.
- We propose VoxelDiffusionCut, a novel framework for non-destructive extraction of a target internal part from a product, based on iterative voxel-based internal structure estimation using a diffusion model and cutting planning.
- We validate the proposed VoxelDiffusionCut through simulation experiments using simple- and complex-shaped models, where the complex-shaped models emulate real products.

2 Related Work

2.1 Studies on Dismantling Automation for Non-Destructive Internal-part Extraction

Many studies have addressed parts extraction at recycling and disposal sites [7, 17, 18]. If product information is available, disassembly can be used to extract internal parts [3–6]. However, such information is rarely disclosed due to concerns about intellectual property. Therefore, product dismantling approaches based on mechanical shredding using shredders or rotating chains, combined with part sorting, have been investigated [19, 20]. Tsunazawa et al. showed that optimizing chain rotation speed and crushing time enables efficient recovery of waste printed circuit boards (PCBs) [21]. Such approaches are effective for material recovery but unsuitable when valuable parts must not be shredded.

To extract parts that are difficult to shred, methods using X-ray imaging, including tomographic reconstruction techniques, have been proposed to visualize internal structures [22] or identify parts for separation [23]. Ueda et al. proposed extracting internal parts by applying impact at the product joints [2]. The optimal impact position is selected using a genetic algorithm based on the battery location obtained from X-ray imaging. However, such X-ray-based approaches assume that internal structures are clearly captured by X-ray equipment, which restricts their applicability to thin products, such as smartphones. For thick products, such as vacuum cleaners, X-ray imaging often lacks sufficient penetration depth to provide detailed internal information.

In contrast, mechanical separation by cutting can be applied regardless of product thickness, but the internal structure remains unknown, which makes the decision of where to cut challenging. Therefore, we explore a new framework for the extraction of the target internal part that combines internal structure estimation by a diffusion model and iterative cutting planning. This approach can be applied even to thick products, since internal structure estimation relies on cutting surfaces obtained through cutting.

2.2 Diffusion Models for 3D Shape Generation

Diffusion models [13–15] are a class of generative models with high expressiveness and flexibility. They have been applied to diverse tasks, such as text-to-image generation [24–26], video editing [27–29] and robot control [30–32].

As an example of 3D shape generation, Luo et al. proposed a diffusion model for point cloud generation [33]. Tang et al. proposed a method that generates 3D shapes from depth images using two diffusion models trained in a hierarchical manner [34]. Nichol et al. reported a method for generating diverse 3D shapes conditioned on text [35], which combines a text-to-image model with a 3D shape-generation model. In addition, diffusion models have been applied to other 3D shape representations, including explicit representations (voxel grids [36] and meshes [37]) and implicit representations (occupancy fields [38] and neural radiance fields [39]). However, these approaches are designed to generate 3D object geometry and do not explicitly represent internal structures composed of multiple parts. Moreover, directly learning 3D shapes is challenging due to the high dimensionality of shape representations [8–10].

In contrast, our method represents 3D shapes as voxels and encodes constituent parts as voxel attributes, enabling internal structure representation. With this shape representation, the model only needs to predict these attributes on a regular grid structure, which makes the learning more tractable. Furthermore, we complete the internal structure using a diffusion model conditioned on the observed cutting surfaces. This approach captures multi-modal predictive uncertainty in unobserved regions, thereby mitigating mode collapse and overconfident predictions. By leveraging the estimated uncertainty during cutting planning, we can help to achieve non-destructive extraction of the target internal part.

3 Preliminary

As preparation for the proposed method, we describe diffusion models (§3.1) and conditional diffusion models (§3.2). Specifically, we introduce Classifier-Free Guidance (CFG) generation [40, 41] as a conditional diffusion model for completing partial observations.

3.1 Denoising Diffusion Probabilistic Models

Diffusion models [13–15] consist of a forward diffusion process, which gradually transforms the original data into Gaussian noise, and a denoising process, which transforms Gaussian noise into samples from the data distribution. Let $\mathbf{x}^0$ denote the original data and $\mathbf{x}^n$ the data after adding noise n times. The forward diffusion process is defined as $q(\mathbf{x}^{1:N}|\mathbf{x}^0) := \prod_{n=1}^N q(\mathbf{x}^n|\mathbf{x}^{n-1})$, $q(\mathbf{x}^n|\mathbf{x}^{n-1}) := \mathcal{N}(\mathbf{x}^n; \sqrt{1 - \beta_n}\mathbf{x}^{n-1}, \beta_n\mathbf{I})$. Here, β_n is the noise scheduler, which controls the noise scale.

The denoising process is represented as a Gaussian distribution, where the mean is given by a model parameterized by θ , which takes the noise data $\mathbf{x}^n$ and the diffusion step n as inputs:

$$p_\theta(\mathbf{x}^{0:N}) := p(\mathbf{x}^N) \prod_{n=1}^N p_\theta(\mathbf{x}^{n-1}|\mathbf{x}^n), \quad (1)$$

$$p_\theta(\mathbf{x}^{n-1}|\mathbf{x}^n) := \mathcal{N}(\mathbf{x}^{n-1}; \boldsymbol{\mu}_\theta(\mathbf{x}^n, n), \boldsymbol{\Sigma}^n), \quad (2)$$

$$p(\mathbf{x}^N) = \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (3)$$

The covariance matrix Σ^n is given by $\Sigma^n = \tilde{\beta}_n \mathbf{I}$ [14], where $\tilde{\beta}_n = \beta_n(1 - \bar{\alpha}_{n-1})/(1 - \bar{\alpha}_n)$, $\bar{\alpha}_n = \prod_{s=1}^n \alpha_s$, and $\alpha_s = 1 - \beta_s$. In the training process, instead of directly optimizing μ_θ , the model learns to predict the noise ϵ from noisy data $\mathbf{x}^n$ using the simplified loss function:

$$\mathcal{L}_\theta = \mathbb{E}_{n, \epsilon, \mathbf{x}^0} [\|\epsilon - \epsilon_\theta(\mathbf{x}^n, n)\|^2], \quad (4)$$

since μ_θ can be expressed in terms of ϵ_θ as $\mu_\theta(\mathbf{x}^n, n) = \frac{1}{\sqrt{\alpha_n}}(\mathbf{x}^n - \frac{1-\alpha_n}{\sqrt{1-\alpha_n}}\epsilon_\theta(\mathbf{x}^n, n))$.

3.2 Conditional Diffusion with Classifier-Free Guidance

Conditional diffusion models [40, 42] generate data $\mathbf{x}^0$ based on conditions $\mathbf{c}$ such as labels, text, or observed information. They are trained by minimizing the conditional loss:

$$\mathcal{L}_\theta^{\text{cond}} = \mathbb{E}_{n, \epsilon, \mathbf{x}^0, \mathbf{c}} [\|\epsilon - \epsilon_\theta(\mathbf{x}^n, n, \mathbf{c})\|^2]. \quad (5)$$

To strengthen conditioning during sampling, classifier guidance [15] has been proposed, which steers the denoising process using gradients from an external classifier. However, classifier guidance requires training an additional classifier and may become unreliable when the conditioning information is weak or ambiguous.

To mitigate this, Classifier-Free Guidance (CFG) [40] has been proposed. CFG enables a single diffusion model to perform both conditional and unconditional generation by randomly dropping the condition $\mathbf{c}$ during training. This prevents overfitting to the condition and allows flexible, diverse generation. During the denoising process, the predicted noise is computed as a linear combination of the conditional $\epsilon_\theta(\mathbf{x}^n, n, \mathbf{c})$ and unconditional $\epsilon_\theta(\mathbf{x}^n, n)$ predictions:

$$\epsilon_{\text{guided}} = (1 + w)\epsilon_\theta(\mathbf{x}^n, n, \mathbf{c}) - w\epsilon_\theta(\mathbf{x}^n, n). \quad (6)$$

Here, w is the guidance scale that controls the conditioning strength.

4 Proposed Method

In this section, we describe the proposed VoxelDiffusionCut framework for non-destructive extraction of the target internal part from a product with an unknown internal structure (§4.1). §4.2 introduces voxel-based 3D shape representation, §4.3 describes conditional diffusion models for internal-structure estimation, and §4.4 presents the cutting action planning method for extracting the target internal part without damage. Video 1 (00:26–01:37) provides an illustrative explanation of the framework components and the iterative inference–planning loop shown in Fig. 2.

4.1 VoxelDiffusionCut Framework

VoxelDiffusionCut consists of three main stages (Fig. 2):

(1) **Data collection** Collect 3D shapes with multiple internal-structure arrangements and types of parts as training data for an internal-structure prediction model that uses a diffusion model.

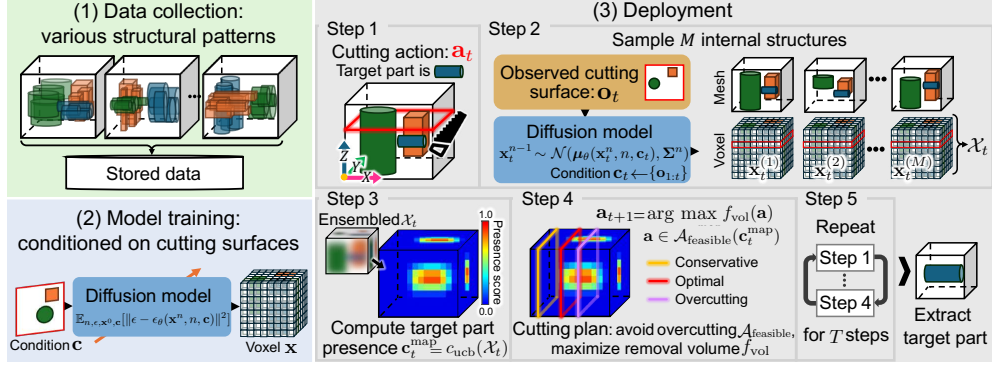


Fig. 2. Framework of VoxelDiffusionCut. The internal structure prediction model is trained using a diffusion model based on data with various location patterns of internal parts. During deployment, the model estimates the internal structure through conditional generation based on the observed cutting surfaces and subsequently plans the cutting positions.

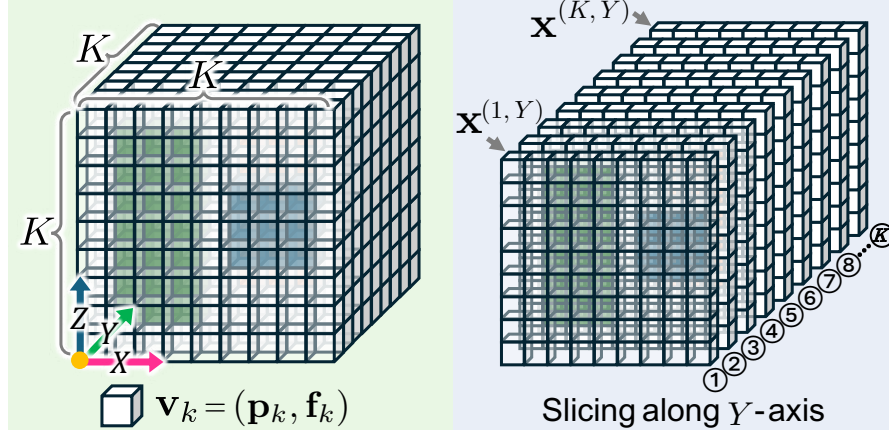
(2) **Model training** Voxelize the collected 3D shapes and encode the constituent parts as voxel attributes. Then train the diffusion model, conditioned on the cutting surfaces, to estimate the internal structure within the voxel space.

(3) **Deployment** Extract the target internal part without damaging it by performing the following steps:

Step 1. Execute the cutting action $\mathbf{a}_t \in \mathcal{A}$, where $t \in \{1, \dots, T\}$ denotes the task step. The initial cutting action is set to avoid damaging the target part, assuming that prior information, such as the product type, is available. **Step 2.** Predict the internal structures $\mathbf{x}_t$ using the diffusion model conditioned on the observed cutting surface $\mathbf{o}_t$. Diffusion models generate data by sampling from a learned distribution, which leads to diverse outputs. Such sample-based diversity has been used as an indicator of predictive uncertainty in robotic applications [43–45]. Following this approach, we also leverage this sample-based uncertainty to inform our cutting plan. Therefore, we predict M internal structures $\mathcal{X}_t = \{\mathbf{x}_t^{(m)}\}_{m=1}^M$. **Step 3.** Compute the presence score map $\mathbf{c}_t^{\text{map}}$ of the target part using $c_{\text{ucb}}(\mathcal{X}_t)$. **Step 4.** Plan the next cutting action $\mathbf{a}_{t+1}$ based on $\mathbf{c}_t^{\text{map}}$ to maximize removable volume while avoiding damage to the target part. **Step 5.** Repeat Steps 1–4 for T task steps, which enables VoxelDiffusionCut to achieve non-destructive extraction of the target internal part.

4.2 3D Shape Representation by Voxelization with Part-Attribute Encoding

We represent 3D shapes as voxels and simplify the training of the internal-structure prediction model by learning the voxel attributes that encode the constituent parts. In the voxel representation, the 3D space is divided along each axis into K uniform cubes, which gives a total of K^3 voxels. Each voxel is assigned attributes (e.g., color, material type) to represent constituent parts of the 3D shape. We denote $\mathbf{p}_k$ as the 3D



(a) Voxel representation

(b) Cutting surface examples

Fig. 3. 3D shape representation by voxelization. (a) Voxel representation with a discrete 3D grid K . (b) Examples of cutting surfaces (slices) extracted along the Y -axis (indexed by j), with each surface denoted as $\mathbf{x}^{(i,j)}$.

coordinates of the k -th voxel and $\mathbf{f}_k$ as the feature vector representing its attributes. A 3D shape is then represented in the voxel space as $\mathcal{V} = \{\mathbf{v}_k\} = \{(\mathbf{p}_k, \mathbf{f}_k)\}$.

Therefore, the observation $\mathbf{o}_t$ obtained by cutting can be interpreted as the cutting surface of the internal structure $\mathbf{x}_t^{(i,j)}$ ($\mathbf{o}_t := \mathbf{x}_t^{(i,j)}$). Here, $j \in \{X, Y, Z\}$ is the cutting axis and $i \in \{1, \dots, K\}$ specifies the cutting position. The cutting action can be represented as $\mathbf{a}_t := [i, j]$. Figure 3 (a) shows a 3D voxel representation, and Fig. 3 (b) illustrates cutting-surface examples obtained by the cutting action.

4.3 Internal Structure Estimation Using Conditional Diffusion Models

We use a conditional diffusion model with CFG to estimate the internal structure $\mathbf{x}_t \in \mathbb{R}^{K \times K \times K \times 3}$ given observations up to task step t , denoted as $\mathbf{x}'_t := \{\mathbf{o}_{1:t}\}$. We define $\mathbf{x}'_t$ and its mask matrix $\mathbf{m}_t \in \mathbb{Z}_2^{K \times K \times K}$ as condition $\mathbf{c}_t := \{\mathbf{x}'_t, \mathbf{m}_t\}$. The denoising step for $\mathbf{x}^{n-1}$ can be expressed as follows, based on Eq. (2):

$$\mathbf{x}_t^{n-1} \sim \mathcal{N}(\mathbf{x}_t^{n-1}; \boldsymbol{\mu}_\theta^{\text{guided}}(\mathbf{x}_t^n, n, \mathbf{c}_t), \boldsymbol{\Sigma}^n). \quad (7)$$

Here, $\boldsymbol{\mu}_\theta^{\text{guided}}(\mathbf{x}_t^n, n, \mathbf{c}_t) := \frac{1}{\sqrt{\alpha_n}}(\mathbf{x}_t^n - \frac{1-\alpha_n}{\sqrt{1-\alpha_n}} \cdot \epsilon_\theta^{\text{guided}})$, and thus $\epsilon_\theta^{\text{guided}}$ is computed as follows according to Eq. (6).

$$\epsilon_\theta^{\text{guided}} = (1 + w)\epsilon_\theta(\mathbf{x}_t^n, n, \mathbf{c}_t) - w\epsilon_\theta(\mathbf{x}_t^n, n). \quad (8)$$

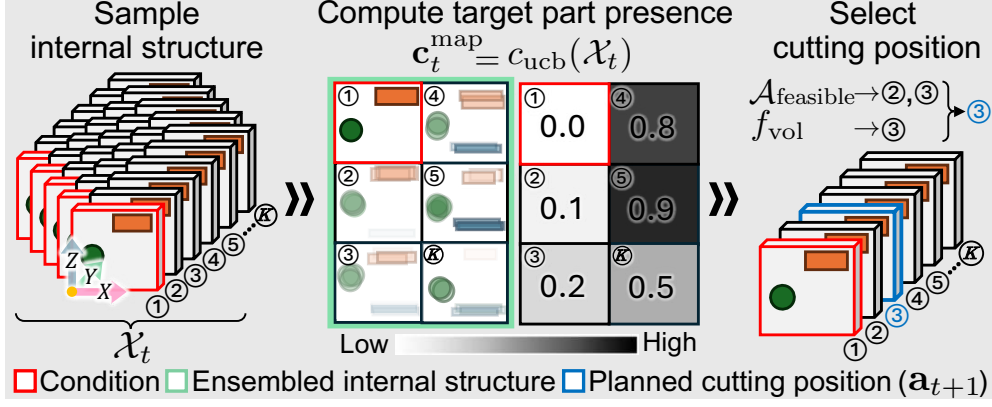


Fig. 4. Cutting action planning along the Y-axis based on predicted internal structure. Values on each cutting surface indicate the presence score of the target internal part. Positions ② and ③ are feasible owing to their low presence scores, and ③ is selected because it maximizes the removable volume.

4.4 Cutting Action Planning

As requirements for planning the next cutting action, it is desirable to (i) avoid damage to the target internal part and (ii) maximize the removable volume. Given the presence score map $\mathbf{c}_t^{\text{map}}$, which represents an estimate of the target part's presence on each cutting surface based on the predicted internal structures $\mathcal{X}_t$, one possible formulation for planning the next cutting action $\mathbf{a}_{t+1}$ is given by

$$\mathbf{a}_{t+1} = \arg \max_{\mathbf{a} \in \mathcal{A}_{\text{feasible}}(\mathbf{c}_t^{\text{map}})} f_{\text{vol}}(\mathbf{a}). \quad (9)$$

Here, $\mathcal{A}_{\text{feasible}}(\cdot)$ is the set of candidate cutting actions for which the presence score map $\mathbf{c}_t^{\text{map}}$ does not exceed the cutting-risk threshold η ($\eta \geq 0$):

$$\mathcal{A}_{\text{feasible}}(\mathbf{c}_t^{\text{map}}) = \{\mathbf{a} \in \mathcal{A} \mid \mathbf{c}_t^{\text{map}}(i, j) \leq \eta\}. \quad (10)$$

$f_{\text{vol}}(\cdot)$ is a function that computes the removal volume corresponding to a cutting position. Therefore, a small η reduces the risk of damaging the target part but limits the removal volume, whereas a large η increases this risk but allows larger removal volume.

In our method, $\mathbf{c}_t^{\text{map}}$ is defined as the mean and standard deviation of $f_s(\mathcal{X}_t)$, where $f_s(\cdot)$ is the part detector, i.e., a binary classifier that indicates whether the target voxel feature $\mathbf{f}^{\text{target}}$ exists on each cutting surface $\{\mathbf{x}_t^{(i,j)}\}^m$:

$$\begin{aligned} \mathbf{c}_t^{\text{map}}(i, j) &= c_{\text{ucb}}(\mathcal{X}_t; i, j), \\ &= \mu_m(f_s(\mathbf{x}_t^{(m)}; i, j)) + \gamma \sigma_m(f_s(\mathbf{x}_t^{(m)}; i, j)), \end{aligned} \quad (11)$$

where $\mu_m(\cdot)$ and $\sigma_m(\cdot)$ compute the mean and standard deviation, respectively, and γ is the weight for the standard deviation². Part detector $f_s(\cdot)$ is defined as

$$f_s(\mathbf{x}_t; i, j) = \mathbb{1}[\mathbf{f}^{\text{target}} \in \mathbf{x}_t^{(i,j)}]. \quad (12)$$

In this work, each cutting surface is represented as a set of voxels. Therefore, the target part’s presence is evaluated by determining if any voxel in the cutting surface corresponds to the target voxel feature $\mathbf{f}^{\text{target}}$. Figure 4 illustrates how the cutting action is planned along the Y -axis.

5 Experiments

To verify the effectiveness of the proposed method, we built a geometric cutting simulator and conducted experiments on simple-shaped models composed of primitive geometries and complex-shaped models that emulate real products. This experiment had two objectives:

- To confirm that the proposed method enables the extraction of a target internal part based on the observed cutting surfaces. In particular, we evaluate the effect of the cutting-risk threshold η , used for cutting planning (§5.3.1, 5.5.1).
- To compare the task performance of the proposed method with those of baseline methods (§5.3.2, 5.5.2).

The iterative cutting process for both the simple- and complex-shaped models is shown in Video 1 (01:38–04:13).

5.1 Common Experimental Setup

5.1.1 Preparation of Benchmark Models

As shown in Fig. 5, the simple-shaped models consist of an external shape (gray) enclosing three cuboids, while the complex-shaped models emulate a real product (a sheet sander) with three components: a battery (blue), a PCB (green), and a motor (red). In the simple-shaped models, Part B was set as the battery and extracted as the target internal part. In these experiments, we assumed that the color of each part could be identified from the observed cutting surfaces, and we used the identified color as voxel feature $\mathbf{f}_k := [R, G, B] \in \mathbb{R}^3$. In real products, analogous part-related cues (which we represent here as color-coded voxel features) can be obtained from raw cutting-surface observations by leveraging modern segmentation models (e.g., Mask R-CNN [48] or SAM [49–51]); see §6 for further discussion.

5.1.2 Simulation Environments

For shape editing and rendering, we extended PyVista [52] and Blender [53]. While cutting errors and observation noise can occur in real environments, the simulation

²While we use a UCB-style score $\mu + \gamma\sigma$ in Eq. (11), the planner is not tied to this particular choice. Any risk function over the Monte Carlo samples $f_s(\cdot)$ can be used to define $\mathbf{c}_t^{\text{map}}$, e.g., a Value at Risk [46], Conditional Value at Risk [47], or a worst-case (max-sample) score. These alternatives trade off conservativeness and efficiency in different ways and can be selected depending on application requirements.

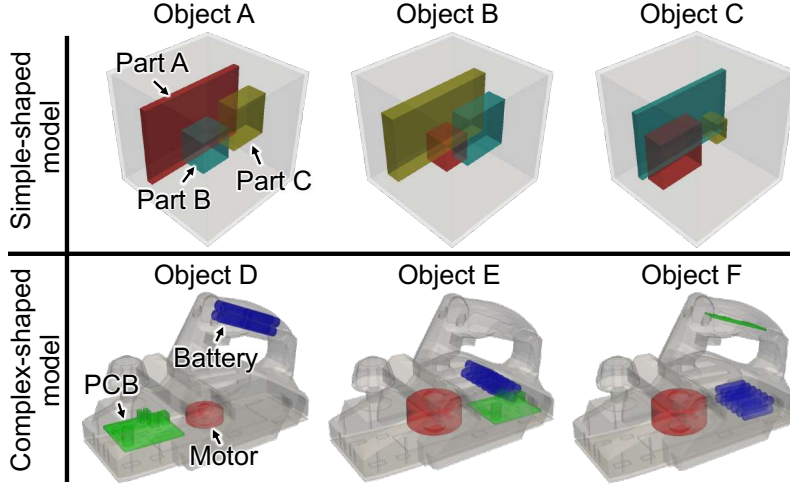


Fig. 5. Object configurations used for evaluation. The battery is defined as the target internal part for extraction. In the simple-shaped model, Part B corresponds to the battery.

experiments assumed planned cuts were executed without error and cutting surfaces were observed accurately. This setting focuses on evaluating the core capability of the proposed framework to estimate internal structures from observed cutting surfaces and to extract the target part by leveraging the estimated uncertainty. At each step, unnecessary regions were assumed to be discarded immediately after cutting and thus became unobservable thereafter.

5.1.3 Baseline Methods

We prepared the following baseline methods.

- **Random:** A method that randomly selects cutting actions at each task step and performs cutting.
- **VAEAC:** A method that predicts internal structures using a CVAE-based generative model [54].
- **PCD-DM:** A method that directly predicts 3D shapes represented as colored point clouds using a diffusion model with a 1D-UNet. While the proposed method outputs $3K^3$ dimensions for voxel colors, **PCD-DM** directly outputs the position $\mathbf{p}$ and color $\mathbf{f}$ of each point, with an output dimensionality of $6K^3$.
- **Proposed-Nocond:** The proposed method without conditioning on the observed cutting surfaces.
- **Proposed-GT:** A method that plans cuts with known internal structures, which is used to evaluate the upper bound of task performance for the proposed method.

Except for **Random**, all baseline methods used the same cutting action planning procedure as the proposed method.

5.1.4 Task Performance Metrics

We prepared three metrics to evaluate task performance, where the volumes were computed from the number of voxels.

- **Cutting Error Volume** [voxels]: Total volume of the target part that was cut erroneously during the task.
- **Part Remaining Rate** [%]: Retention rate of the target part, defined as (volume of target part after task/volume of target part before task) $\times 100$.
- **Part Occupancy Rate** [%]: Proportion of the target part volume to the total volume of the shape after the task, defined as (volume of target part after task/total volume of shape after task) $\times 100$.

5.1.5 Cutting Action Planning Settings

In this experiment, since color information was used as the voxel feature, we computed the part detector $f_s(\mathcal{X}_t)$ given by Eq. (12) for each cutting surface $\mathbf{x}_t^{(i,j)}$ to evaluate the target part’s presence, as follows:

$$f_s(\mathbf{x}_t; i, j) = \mathbb{1}[\exists d \text{ s.t. } \mathbf{f}_{\min}^{\text{target}} \leq \mathbf{x}_t^{(i,j)}[d] \leq \mathbf{f}_{\max}^{\text{target}}], \quad (13)$$

where, $\mathbf{x}_t^{(i,j)}[d]$ denotes the color value of the d -th voxel on the cutting surface $d \in \{1, \dots, K^2\}$ and $\mathbf{f}_{\min}^{\text{target}}, \mathbf{f}_{\max}^{\text{target}}$ represent the range of color values of the target part.

In Eq. (11), we set the scale parameter $\gamma = 1.0$ to equally evaluate the mean and the standard deviation. As a simple implementation, we solve the optimization in Eq. (9) via an exhaustive search over all cutting surfaces. The number of task steps T was set to 8.

5.2 Experimental Setup of Simple-Shaped Model

We prepared five internal-part arrangement distributions and generated 20K samples from each to train the conditional diffusion model. Each distribution had different part sizes and relative spatial relationships (reference arrangements), and the final placement was sampled from a uniform distribution centered on the reference arrangement. The internal structures used for evaluation (Fig. 5) were newly sampled from the same three arrangement distributions used in training. This evaluation follows a standard independent and identically distributed (i.i.d.) setting in statistical machine learning [55], and aims to validate whether the learned model can represent the multi-modal internal-part arrangements of the training distribution.

3D shapes were voxelized at $K = 16$ and used to train the diffusion model with a U-Net architecture for 100K steps (batch size 192, $N = 1000$ diffusion steps). During training, the diffusion model is conditioned on randomly generated masks [54, 56] to cover diverse observation patterns of cutting surfaces. Sampling was performed with a guidance scale of $w = 0.2$ using $M = 32$ samples. Training the diffusion model took approximately 16.4 hours on NVIDIA GeForce RTX 4090. At inference time, we used DDIM sampling [57] with 20 denoising steps to accelerate generation; on average,

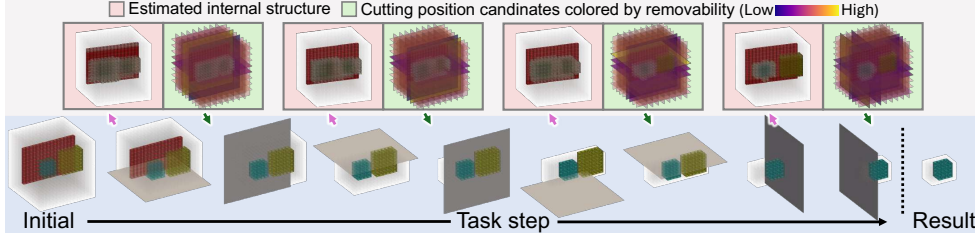


Fig. 6. Examples of cutting process transitions by the proposed VoxelDiffusionCut with the simple-shaped model (Object A). VoxelDiffusionCut plans cutting positions based on the estimated internal structure to maximize the removable volume while non-destructively extracting the target internal part.

Table 1. Task performance comparison under different cutting-risk thresholds for the simple-shaped model (Object A)

η	Cut Err. Vol. $\downarrow$	Part Remain. Rate $\uparrow$	Part Occ. Rate $\uparrow$
0.0	0.0 ± 0.0	100.0 ± 0.0	7.71 ± 0.57
0.5	0.0 ± 0.0	100.0 ± 0.0	27.17 ± 2.66
1.0	188.0 ± 78.59	25.0 ± 28.87	2.84 ± 3.33

producing $M = 32$ conditional samples for one planning step took 0.7 seconds on the same hardware.

5.3 Results of Simple-shaped Models

5.3.1 Task Performance Comparison with Different Cutting-Risk Thresholds

TABLE 1 shows the task performance of Object A evaluated six times with different cutting-risk thresholds η . The results indicate that with $\eta = 0.0$, erroneous cuts are completely avoided, but the part occupancy rate is low, reflecting a conservative cutting plan. With $\eta = 1.0$, the part occupancy rate is high, but the cutting error volume is large. With $\eta = 0.5$, erroneous cuts are avoided while maintaining a high part occupancy rate. Figure 6 shows the cutting process of Object A ($\eta = 0.5$), which was planned based on the estimated internal structure.

These results show that the proposed method achieves efficient removal of unnecessary regions without damaging the target part through the adjustment of η . In the following experiments, we used $\eta = 0.5$, which produced a balanced performance.

5.3.2 Task Performance Comparison with Baseline Methods

TABLE 2 shows the task performance of each method evaluated six times. Figure 7 visualizes examples of the extracted shapes. **VAEAC** and **PCD-DM** suffer from erroneous cuts, leading to a low part remaining rate. At the same time, these erroneous cuts explain why their part occupancy rate occasionally appears high: Cutting into

Table 2. Task performance comparison of simple-shaped models. In the part occupancy rate column, boldface figures indicate the highest value among the methods without cutting error volume.

(a) Object A

Method	Cut Err. Vol. ↓	Part Remain. Rate ↑	Part Occ. Rate ↑
Random	148.67 ±106.57	42.36 ±21.57	2.5 ±1.16
VAEAC	84.0 ±0.0	75.0 ±0.0	60.0 ±0.0
PCD-DM	66.0 ±30.0	50.0 ±25.0	42.25 ±8.7
Proposed-Nocond	0.0 ±0.0	100.0 ±0.0	8.88 ±1.56
Proposed	0.0 ±0.0	100.0 ±0.0	27.17 ±2.66
Proposed-GT	0.0 ±0.0	100.0 ±0.0	48.0 ±0.0

(b) Object B

Method	Cut Err. Vol. ↓	Part Remain. Rate ↑	Part Occ. Rate ↑
Random	490.5 ±157.2	30.69 ±15.54	4.59 ±2.54
VAEAC	252.0 ±0.0	71.43 ±0.0	75.0 ±0.0
PCD-DM	304.5 ±109.62	41.67 ±20.97	51.53 ±5.24
Proposed-Nocond	0.0 ±0.0	100.0 ±0.0	21.87 ±1.4
Proposed	0.0 ±0.0	100.0 ±0.0	48.61 ±4.35
Proposed-GT	0.0 ±0.0	100.0 ±0.0	58.33 ±0.0

(c) Object C

Method	Cut Err. Vol. ↓	Part Remain. Rate ↑	Part Occ. Rate ↑
Random	558.67 ±175.63	21.07 ±15.92	5.75 ±4.73
VAEAC	35.0 ±10.69	90.0 ±0.0	21.87 ±3.09
PCD-DM	291.67 ±103.01	65.0 ±11.18	31.35 ±12.48
Proposed-Nocond	767.0 ±309.49	0.0 ±0.0	0.0 ±0.0
Proposed	0.0 ±0.0	100.0 ±0.0	26.52 ±0.0
Proposed-GT	0.0 ±0.0	100.0 ±0.0	39.77 ±0.0

the target part increased the observed information on the internal structure and thus improved inference accuracy. Compared with **Proposed-Nocond**, which is not conditioned on the observed cutting surfaces, the proposed method achieves a consistently higher part occupancy rate by explicitly conditioning on them.

These results indicate that, for multiple evaluated shapes with different internal structure configurations, the proposed method produces cutting plans that reduce erroneous cuts of the target part while maintaining a higher part occupancy rate than the baseline approaches.

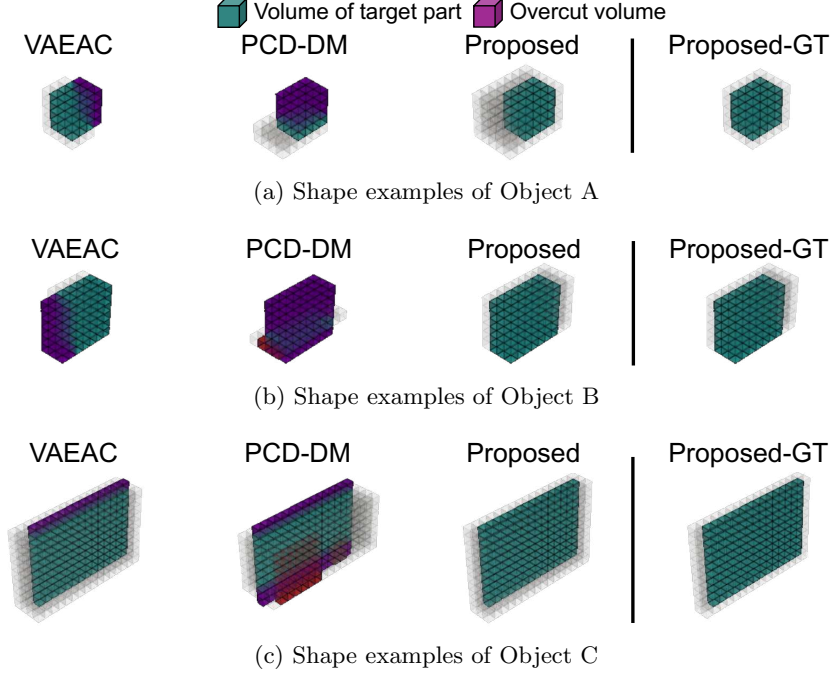


Fig. 7. Rendered objects after processing by each method for the simple-shaped models

5.4 Experimental Setup of Complex-shaped Models

We prepared six internal-part arrangement distributions and generated 3K samples from each to train the conditional diffusion model. Each distribution had different part sizes and relative spatial relationships, as with the simple-shaped models. The final placement was sampled from a uniform distribution centered on the reference arrangement. The internal structures used for evaluation (Fig. 5) were newly sampled from the same arrangement distributions used in training, following the same evaluation setting as for the simple-shaped models.

3D shapes were voxelized at $K = 46$ and used to train the diffusion model with a U-Net architecture for 100K steps (batch size 90, $N = 1000$ diffusion steps). During training, the diffusion model is conditioned on randomly generated masks [54, 56] to cover diverse observation patterns of cutting surfaces. Sampling was performed with a guidance scale of $w = 0.2$ using $M = 32$ samples. Training the diffusion model for 100K steps took approximately 5.3 days on (NVIDIA H100 NVL). Inference was performed with DDIM sampling [57] as in the simple-shaped setup; producing $M = 32$ conditional samples with 20 denoising steps for one planning step took 26.7 seconds on average on the same hardware.



Fig. 8. Examples of cutting process transitions by the proposed VoxelDiffusionCut with the complex-shaped model (Object D)

Table 3. Task performance comparison under different cutting-risk thresholds for the complex-shaped model (Object D)

η	Cut Err. Vol. $\downarrow$	Part Remain. Rate $\uparrow$	Part Occ. Rate $\uparrow$
0.0	0.0 ± 0.0	100.0 ± 0.0	3.94 ± 2.14
0.5	0.0 ± 0.0	100.0 ± 0.0	33.14 ± 6.25
1.0	38.33 ± 55.81	88.89 ± 15.71	20.0 ± 8.26

5.5 Results of Complex-shaped Models

5.5.1 Task Performance Comparison with Different Cutting-Risk Thresholds

TABLE 3 shows the task performance of Object D evaluated six times with different cutting-risk thresholds η . These results show that the proposed method achieves efficient removal of unnecessary regions without damaging the target part through the adjustment of η , similar to the simple-shaped models. Figure 8 visualizes the cutting process of Object D ($\eta = 0.5$).

5.5.2 Task Performance Comparison with Baseline Methods

TABLE 4 shows the task performance of each method evaluated six times. Figure 9 visualizes examples of the extracted shapes. **PCD-DM** avoids erroneous cuts but exhibits a low part occupancy rate. This suggests that **PCD-DM**, which directly generates 3D shapes as colored point clouds, struggles to plan effective cutting actions for extracting the target part based on the estimated internal structures in high-dimensional and complex shapes.

In Object F, **VAEAC** produced erroneous cuts, whereas **Proposed** avoided them but did not achieve battery-only extraction. Figure 10 shows the presence score map of the estimated internal structure at task step $t = 8$. The figure shows that the proposed method, which uses a conditional diffusion model, provided high-confidence internal structure estimation for Objects D and E while capturing uncertainty for Object F. In contrast, **VAEAC** had errors in estimating the internal structure across all objects. These results indicate that the proposed method provides uncertainty-aware estimation and cutting plans, even when the internal structure cannot be represented as a single distribution from the observed cutting surfaces. This characteristic is essential at recycling and disposal sites, where parts such as batteries must be extracted without damage for safety.

Table 4. Task performance comparison of complex-shaped models. In the part occupancy rate column, boldface figures indicate the highest value among the methods without cutting error volume.

(a) Object D

Method	Cut Err. Vol. ↓	Part Remain. Rate ↑	Part Occ. Rate ↑
Random	332.17 ±252.73	57.0 ±25.49	0.19 ±0.09
VAEAC	0.0 ±0.0	100.0 ±0.0	0.24 ±0.0
PCD-DM	0.0 ±0.0	100.0 ±0.0	0.24 ±0.0
Proposed-Nocond	0.0 ±0.0	100.0 ±0.0	4.3 ±0.68
Proposed	0.0 ±0.0	100.0 ±0.0	33.14 ±6.25
Proposed-GT	0.0 ±0.0	100.0 ±0.0	35.94 ±0.0

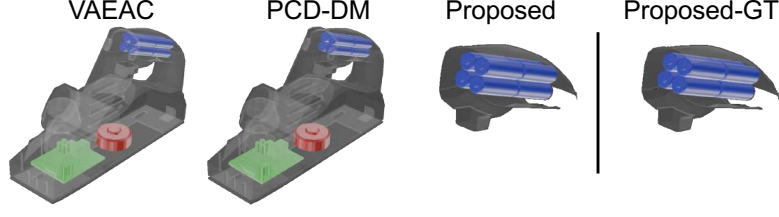
(b) Object E

Method	Cut Err. Vol. ↓	Part Remain. Rate ↑	Part Occ. Rate ↑
Random	275.33 ±128.77	70.71 ±13.66	0.43 ±0.09
VAEAC	25.0 ±0.0	89.18 ±0.0	61.31 ±0.0
PCD-DM	0.0 ±0.0	100.0 ±0.0	0.41 ±0.0
Proposed-Nocond	26.0 ±58.14	94.37 ±12.58	5.4 ±1.52
Proposed	0.0 ±0.0	100.0 ±0.0	41.39 ±9.03
Proposed-GT	0.0 ±0.0	100.0 ±0.0	55.0 ±0.0

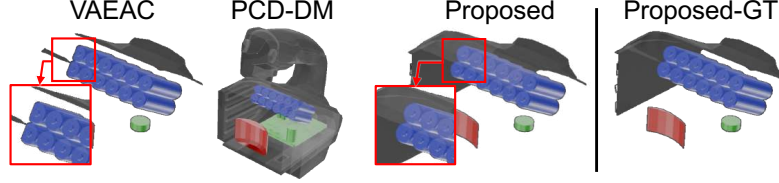
(c) Object F

Method	Cut Err. Vol. ↓	Part Remain. Rate ↑	Part Occ. Rate ↑
Random	261.5 ±68.83	57.78 ±8.64	0.2 ±0.03
VAEAC	75.0 ±0.0	88.89 ±0.0	23.44 ±0.0
PCD-DM	0.0 ±0.0	100.0 ±0.0	0.25 ±0.0
Proposed-Nocond	0.0 ±0.0	100.0 ±0.0	3.35 ±0.96
Proposed	0.0 ±0.0	100.0 ±0.0	5.47 ±0.0
Proposed-GT	0.0 ±0.0	100.0 ±0.0	56.25 ±0.0

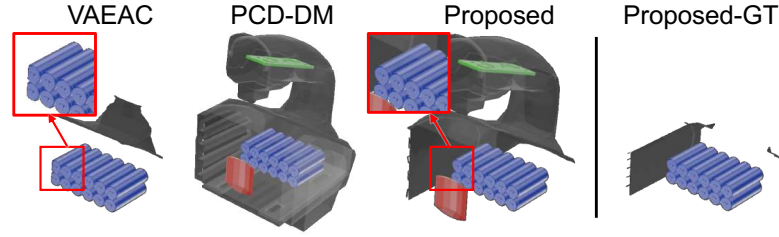
These results confirm that, even for high-dimensional, complex-shaped models, the proposed method provides cutting plans that reduce erroneous cuts under high uncertainty and achieve a high part occupancy rate when the internal structure can be estimated.



(a) Shape examples of Object D



(b) Shape examples of Object E



(c) Shape examples of Object F

Fig. 9. Rendered objects after processing by each method for complex-shaped models. The areas enlarged in red frames show where the baseline method erroneously cut the target internal part.

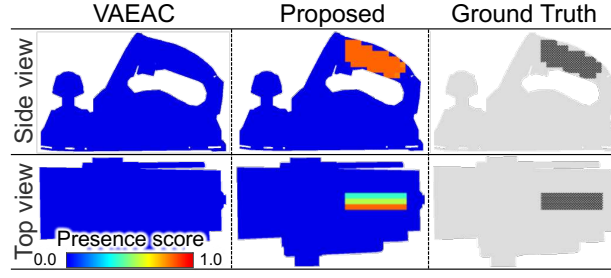
6 Discussion

6.1 Initial Cutting Selection

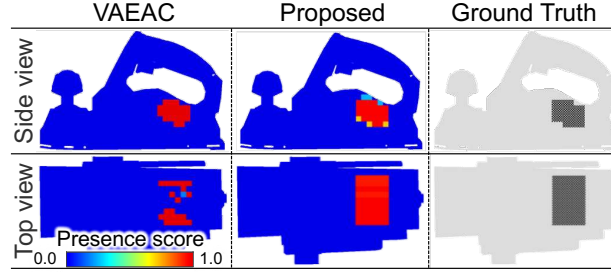
A limitation of the proposed method is the need to predefine the cutting position at the initial step. Since our method estimates internal structure from observed cutting surfaces, it cannot be applied when such observations are unavailable. To address this, we are investigating an extension that leverages external shape and cutting-surface observations through conditioning on information such as product type and appearance. This extension is expected to permit automatic selection of the initial cutting position from external information.

6.2 Real-World Deployment

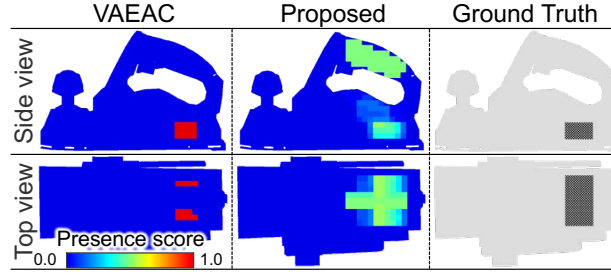
Applying the proposed method to real environments is an interesting direction for future work. One additional challenge in such settings is the observation noise of



(a) Object D



(b) Object E



(c) Object F

Fig. 10. Comparison of presence score maps for target internal part locations in complex-shaped models at task step $t=8$

cutting surfaces, e.g., irregular textures, debris, occlusions, or illumination changes. Recent advances in object recognition and segmentation models have demonstrated robustness to noisy observations, ranging from conventional instance segmentation models such as Mask R-CNN [48] to more recent foundation models such as the Segment Anything Model (SAM) [49–51]. Therefore, incorporating such models to identify internal part types from observed cutting surfaces may enable the proposed loop to operate under realistic visual variability, by using generic cues such as geometry, texture, and material-dependent appearance instead of assuming pre-defined color labels.

Another challenge is related to the execution of planned cutting actions. Physical constraints, including fixed product posture, limited manipulator workspace, and deviations caused by cutting resistance forces, can lead to discrepancies between planned and executed cuts. Such discrepancies are particularly relevant for contact-rich cutting tools (e.g., saw- or grinding-based operations), where tool deflection and reaction forces may introduce large cutting errors. In grinding operations, automation methods have been proposed to compensate for errors due to grinding resistance [58]. Therefore, it is important to plan cutting positions that consider both the estimated internal structure and such physical constraints.

At the same time, it is worth noting that the idealized assumptions used in this paper are not necessarily far from certain modern industrial cutting processes. For example, abrasive waterjet cutting [59] can produce relatively clean and planar cuts under appropriate fixturing and process conditions, reducing surface irregularities and geometric deviations compared to contact-rich cutting tools. In addition, diamond wire sawing [60] has been reported as an effective separation process in photovoltaic module recycling, enabling high-precision cutting with reduced kerf loss under controlled conditions. In such settings, the gap between the nominal cut plane and the realized cut, as well as the ambiguity in the observed cutting surface, can be substantially smaller, making our simulation setting a reasonable first approximation. A systematic evaluation under realistic noise and execution-error models, together with process-specific constraints and cost considerations, remains an important direction for future work.

7 Conclusion

We propose VoxelDiffusionCut, a framework for non-destructive extraction of the target internal part from a product with an unknown internal structure using observations on the cutting surfaces. The problem was formulated as a generative task, in which a diffusion model was applied to complete the voxel representation based on the observed cutting surfaces. Experimental results in simulation suggest that the proposed method can estimate internal structures from observed cutting surfaces and enable non-destructive extraction of the target internal part by leveraging the estimated uncertainty.

Supplementary information. We have provided the supplementary video material.

Acknowledgements. This work is based on results obtained from a project, JPNP23002, commissioned by the New Energy and Industrial Technology Development Organization (NEDO).

Authors contributions. Takumi Hachimine: Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization. Yuhwan Kwon: Resources, Visualization, Writing - Review & Editing. Cheng-Yu Kuo: Software, Visualization, Writing - Review & Editing. Tomoya Yamanokuchi: Project administration, Writing - Review & Editing. Takamitsu Matsubara: Conceptualization, Funding acquisition, Project administration, Supervision, Writing - Review & Editing.

Data availability. The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests. The authors declare that they have no competing interests.

Ethical and informed consent for data used. This article does not contain any studies with human participants or animals.

References

- [1] Das, S.K., Ellamparuthy, G., Kundu, T., Angadi, S.I., Rath, S.S.: A comprehensive review of the mechanical separation of waste printed circuit boards. *Process Safety and Environmental Protection* **187**, 221–239 (2024) <https://doi.org/10.1016/j.psep.2024.04.090>
- [2] Ueda, T., Fukusawa, H., Nakagawa, Y., Nagano, K., Sunahara, N., Yamada, H., Fujisawa, J., Yamada, M., Koyanaka, S., Oki, T.: Automatic high-speed smart-phone disassembly system. *Journal of Cleaner Production* **434**, 139928 (2024) <https://doi.org/10.1016/j.jclepro.2023.139928>
- [3] Kiyokawa, T., Harada, K., Wan, W., Ishikura, T., Miyaji, N., Matsuda, G.: Many-objective-optimized semi-automated robotic disassembly sequences. *Robotics and Autonomous Systems* **197**, 105301 (2026) <https://doi.org/10.1016/j.robot.2025.105301>
- [4] Laili, Y., Li, X., Wang, Y., Ren, L., Wang, X.: Robotic disassembly sequence planning with backup actions. *IEEE Transactions on Automation Science and Engineering* **19**(3), 2095–2107 (2022) <https://doi.org/10.1109/TASE.2021.3072663>
- [5] Zhao, X., Li, C., Tang, Y., Cui, J.: Reinforcement learning-based selective disassembly sequence planning for the end-of-life products with structure uncertainty. *IEEE Robotics and Automation Letters* **6**(4), 7807–7814 (2021) <https://doi.org/10.1109/LRA.2021.3098248>
- [6] Suárez-Hernández, A., Torras, C., Alenyà, G.: Practical resolution methods for mdps in robotics exemplified with disassembly planning. *IEEE Robotics and Automation Letters* **4**(3), 2282–2288 (2019) <https://doi.org/10.1109/LRA.2019.2901905>
- [7] Kiyokawa, T., Takamatsu, J., Koyanaka, S.: Challenges for future robotic sorters of mixed industrial waste: A survey. *IEEE Transactions on Automation Science and Engineering* **21**(1), 1023–1040 (2024) <https://doi.org/10.1109/TASE.2022.3221969>

- [8] Qi, C.R., Su, H., Mo, K., Guibas, L.J.: Pointnet: Deep learning on point sets for 3d classification and segmentation. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 652–660 (2017). <https://doi.org/10.1109/CVPR.2017.16>
- [9] Wang, Y., Sun, Y., Liu, Z., Sarma, S.E., Bronstein, M.M., Solomon, J.M.: Dynamic graph cnn for learning on point clouds. *ACM Transactions on Graphics* **38**(5), 1–12 (2019) <https://doi.org/10.1145/3326362>
- [10] Hanocka, R., Hertz, A., Fish, N., Giryas, R., Fleishman, S., Cohen-Or, D.: Meshcnn: a network with an edge. *ACM Transactions on Graphics* **38**(4), 1–12 (2019) <https://doi.org/10.1145/3306346.3322959>
- [11] Sohn, K., Lee, H., Yan, X.: Learning structured output representation using deep conditional generative models. In: *Advances in Neural Information Processing Systems*, vol. 28 (2015)
- [12] Mirza, M., Osindero, S.: Conditional generative adversarial nets. *arXiv preprint arXiv:1411.1784* (2014) <https://doi.org/10.48550/arXiv.1411.1784>
- [13] Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., Ganguli, S.: Deep unsupervised learning using nonequilibrium thermodynamics. In: *International Conference on Machine Learning*, vol. 37, pp. 2256–2265 (2015)
- [14] Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. In: *Advances in Neural Information Processing Systems*, vol. 33, pp. 6840–6851 (2020)
- [15] Dhariwal, P., Nichol, A.Q.: Diffusion models beat gans on image synthesis. In: *Advances in Neural Information Processing Systems*, vol. 34, pp. 8780–8794 (2021)
- [16] Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., Van Gool, L.: Repaint: Inpainting using denoising diffusion probabilistic models. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11461–11471 (2022). <https://doi.org/10.1109/CVPR52688.2022.01117>
- [17] Mishra, K., Siwal, S.S., Thakur, V.K.: E-waste recycling and utilization: A review of current technologies and future perspectives. *Current Opinion in Green and Sustainable Chemistry* **47**, 100900 (2024) <https://doi.org/10.1016/j.cogsc.2024.100900>
- [18] Hathaway, J., Rastegarpanah, A., Stolkin, R.: Learning robotic milling strategies based on passive variable operational space interaction control. *IEEE Transactions on Automation Science and Engineering* **21**(3), 3435–3448 (2024) <https://doi.org/10.1109/TASE.2023.3279718>
- [19] Lu, Y., Yang, B., Gao, Y., Xu, Z.: An automatic sorting system for electronic

components detached from waste printed circuit boards. *Waste Management* **137**, 1–8 (2022) <https://doi.org/10.1016/j.wasman.2021.10.016>

- [20] Han, S.D., Huang, B., Ding, S., Song, C., Feng, S.W., Xu, M., Lin, H., Zou, Q., Boularias, A., Yu, J.: Toward fully automated metal recycling using computer vision and non-prehensile manipulation. In: *International Conference on Automation Science and Engineering*, pp. 891–898 (2021). <https://doi.org/10.1109/CASE49439.2021.9551431>
- [21] Tsunazawa, Y., Hisatomi, S., Murakami, S., Tokoro, C.: Investigation and evaluation of the detachment of printed circuit boards from waste appliances for effective recycling. *Waste Management* **78**, 474–482 (2018) <https://doi.org/10.1016/j.wasman.2018.06.024>
- [22] Li, Y., Wu, T., Han, Y., Chen, P.: Recognition of incorrect assembly of internal components by x-ray ct and deep learning. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* **920**, 88–94 (2019) <https://doi.org/10.1016/j.nima.2018.12.031>
- [23] Sterkens, W., Diaz-Romero, D., Goedemé, T., Dewulf, W., Peeters, J.R.: Detection and recognition of batteries on x-ray images of waste electrical and electronic equipment using deep learning. *Resources, Conservation and Recycling* **168**, 105246 (2021) <https://doi.org/10.1016/j.resconrec.2020.105246>
- [24] Ramesh, A., Dhariwal, P., Nichol, A., Chu, C., Chen, M.: Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125* (2022) <https://doi.org/10.48550/arXiv.2204.06125>
- [25] Saharia, C., Chan, W., Saxena, S., Li, L., Whang, J., Denton, E.L., Ghasemipour, K., Gontijo Lopes, R., Karagol Ayan, B., Salimans, T., *et al.*: Photorealistic text-to-image diffusion models with deep language understanding. *Advances in Neural Information Processing Systems* **35**, 36479–36494 (2022)
- [26] Zhao, S., Chen, D., Chen, Y.-C., Bao, J., Hao, S., Yuan, L., Wong, K.-Y.K.: Uni-controlnet: All-in-one control to text-to-image diffusion models. *Advances in Neural Information Processing Systems* **36**, 11127–11150 (2023)
- [27] Cohen, N., Kulikov, V., Kleiner, M., Huberman-Spiegelglas, I., Michaeli, T.: Slicedit: Zero-shot video editing with text-to-image diffusion models using spatio-temporal slices. In: *International Conference on Machine Learning*, vol. 235, pp. 9109–9137 (2024)
- [28] Kara, O., Kurtkaya, B., Yesiltepe, H., Rehğ, J.M., Yanardag, P.: Rave: Randomized noise shuffling for fast and consistent video editing with diffusion models. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6507–6516 (2024). <https://doi.org/10.1109/CVPR52733.2024.00622>

- [29] Zhang, Z., Li, B., Nie, X., Han, C., Guo, T., Liu, L.: Towards consistent video editing with text-to-image diffusion models. *Advances in Neural Information Processing Systems* **36**, 58508–58519 (2023)
- [30] Janner, M., Du, Y., Tenenbaum, J., Levine, S.: Planning with diffusion for flexible behavior synthesis. In: *International Conference on Machine Learning*, vol. 162, pp. 9902–9915 (2022)
- [31] Mishra, U., Xue, S., Chen, Y., Xu, D.: Generative skill chaining: Long-horizon skill planning with diffusion models. In: *Conference on Robot Learning*, pp. 2905–2925 (2023)
- [32] Ze, Y., Zhang, G., Zhang, K., Hu, C., Wang, M., Xu, H.: 3d diffusion policy: Generalizable visuomotor policy learning via simple 3d representations. In: *Robotics: Science and Systems* (2024)
- [33] Luo, S., Hu, W.: Diffusion probabilistic models for 3d point cloud generation. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2836–2844 (2021). <https://doi.org/10.1109/CVPR46437.2021.00286>
- [34] Zeng, X., Vahdat, A., Williams, F., Gojcic, Z., Litany, O., Fidler, S., Kreis, K.: Lion: Latent point diffusion models for 3d shape generation. In: *Advances in Neural Information Processing Systems*, vol. 35, pp. 10021–10039 (2022)
- [35] Nichol, A., Jun, H., Dhariwal, P., Mishkin, P., Chen, M.: Point-e: A system for generating 3d point clouds from complex prompts. *arXiv preprint arXiv:2212.08751* (2022) <https://doi.org/10.48550/arXiv.2212.08751>
- [36] Ren, X., Huang, J., Zeng, X., Museth, K., Fidler, S., Williams, F.: Xcube: Large-scale 3d generative modeling using sparse voxel hierarchies. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4209–4219 (2024). <https://doi.org/10.1109/CVPR52733.2024.00403>
- [37] Liu, Z., Feng, Y., Black, M.J., Nowrouzezahrai, D., Paull, L., Liu, W.: Meshdiffusion: Score-based generative 3d mesh modeling. In: *International Conference on Learning Representations* (2023)
- [38] Li, Y., Luo, F., Xiao, C.: Diffusion-fof: Single-view clothed human reconstruction via diffusion-based fourier occupancy field. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9525–9534 (2024). <https://doi.org/10.1109/CVPR52733.2024.00910>
- [39] Wynn, J., Turmukhambetov, D.: Diffusionerf: Regularizing neural radiance fields with denoising diffusion models. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4180–4189 (2023). <https://doi.org/10.1109/CVPR52729.2023.00407>

- [40] Ho, J., Salimans, T.: Classifier-free diffusion guidance. In: NeurIPS Workshop on Deep Generative Models and Downstream Applications (2021)
- [41] Xie, S., Zhang, Z., Lin, Z., Hinz, T., Zhang, K.: Smartbrush: Text and shape guided object inpainting with diffusion model. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 22428–22437 (2023). <https://doi.org/10.1109/CVPR52729.2023.02148>
- [42] Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10684–10695 (2022). <https://doi.org/10.1109/CVPR52688.2022.01042>
- [43] Kumar, A., Power, T., Yang, F., Marinovic, S.A., Iba, S., Zarrin, R.S., Berenson, D.: Diffusion-informed probabilistic contact search for multi-finger manipulation. In: IEEE International Conference on Robotics and Automation, pp. 14277–14283 (2025). <https://doi.org/10.1109/ICRA55743.2025.11127844>
- [44] Lee, S.-W., Kang, X., Kuo, Y.-L.: Diff-dagger: Uncertainty estimation with diffusion policy for robotic manipulation. In: IEEE International Conference on Robotics and Automation, pp. 4845–4852 (2025). <https://doi.org/10.1109/ICRA55743.2025.11127730>
- [45] Lee, K., Kim, S., Choi, J.: Refining diffusion planner for reliable behavior synthesis by automatic detection of infeasible plans. In: Advances in Neural Information Processing Systems, vol. 36, pp. 24223–24246 (2023)
- [46] J.P. Morgan, Reuters: Riskmetrics™—technical document. Technical report, Morgan Guaranty Trust Company, New York (December 1996). Fourth Edition. Dated December 17, 1996. <https://www.msri.com/documents/10199/5915b101-4206-4ba0-ae2-3449d5c7e95a>
- [47] Stan, U.: Conditional value-at-risk: optimization algorithms and applications. In: IEEE/IAFE/INFORMS Conference on Computational Intelligence for Financial Engineering, pp. 49–57 (2000). <https://doi.org/10.1109/CIFER.2000.844598>
- [48] He, K., Gkioxari, G., Dollár, P., Girshick, R.: Mask r-cnn. In: IEEE/CVF International Conference on Computer Vision, pp. 2980–2988 (2017). <https://doi.org/10.1109/ICCV.2017.322>
- [49] Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.-Y., Dollar, P., Girshick, R.: Segment anything. In: IEEE/CVF International Conference on Computer Vision, pp. 4015–4026 (2023). <https://doi.org/10.1109/ICCV51070.2023.00371>
- [50] Ravi, N., Gabeur, V., Hu, Y., Hu, R., Ryali, C., Ma, T., Khedr, H., Rädle, R., Rolland, C., Gustafson, L., Mintun, E., Pan, J., Alwala, K.V., Carion, N., Wu, C.,

Girshick, R.B., Dollár, P., Feichtenhofer, C.: SAM 2: Segment anything in images and videos. In: International Conference on Learning Representations (2025)

- [51] Carion, N., Gustafson, L., Hu, Y.-T., Debnath, S., Hu, R., Coll-Vinent, D.S., Ryali, C., Alwala, K.V., Khedr, H., Huang, A., Lei, J., Ma, T., Guo, B., Kalla, A., Marks, M., Greer, J., Wang, M., Sun, P., Rädle, R., Afouras, T., Mavroudi, E., Xu, K., Wu, T.-H., Zhou, Y., Momeni, L., HAZRA, R., Ding, S., Vaze, S., Porcher, F., Li, F., Li, S., Kamath, A., Cheng, H.K., Dollar, P., Ravi, N., Saenko, K., Zhang, P., Feichtenhofer, C.: SAM 3: Segment anything with concepts. In: International Conference on Learning Representations (2026)
- [52] Sullivan, C.B., Kaszynski, A.: PyVista: 3d plotting and mesh analysis through a streamlined interface for the visualization toolkit (VTK). *Journal of Open Source Software* **4**, 1450 (2019) <https://doi.org/10.21105/joss.01450>
- [53] Community, B.O.: Blender - a 3D Modelling and Rendering Package. Blender Foundation, (2018). Blender Foundation
- [54] Ivanov, O., Figurnov, M., Vetrov, D.: Variational autoencoder with arbitrary conditioning. In: International Conference on Learning Representations (2019)
- [55] Bishop, C.M.: Pattern Recognition and Machine Learning (Information Science and Statistics). Springer, Berlin, Heidelberg (2006)
- [56] Pathak, D., Krähenbühl, P., Donahue, J., Darrell, T., Efros, A.A.: Context encoders: Feature learning by inpainting. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2536–2544 (2016) <https://doi.org/10.1109/CVPR.2016.278>
- [57] Song, J., Meng, C., Ermon, S.: Denoising diffusion implicit models. In: International Conference on Learning Representations (2021)
- [58] Hachimine, T., Morimoto, J., Matsubara, T.: Learning to shape by grinding: Cutting-surface-aware model-based reinforcement learning. *IEEE Robotics and Automation Letters* **8**(10), 6235–6242 (2023) <https://doi.org/10.1109/LRA.2023.3303721>
- [59] Yang, M., Liu, H., Ye, B., Qian, W.: Recycling of printed circuit boards by abrasive waterjet cutting. *Process Safety and Environmental Protection* **148**, 805–812 (2021) <https://doi.org/10.1016/j.psep.2021.01.052>
- [60] Deng, R., Zhuo, Y., Shen, Y.: Recent progress in silicon photovoltaic module recycling processes. *Resources, Conservation and Recycling* **187**, 106612 (2022) <https://doi.org/10.1016/j.resconrec.2022.106612>

```
1
2
3
4 This is pdfTeX, Version 3.141592653-2.6-1.40.26 (TeX Live 2024)
5 (preloaded format=pdflatex 2024.8.2)  4 MAR 2026 06:45
6 entering extended mode
7   restricted \writel8 enabled.
8   %&-line parsing enabled.
9 **root.tex
10 (./root.tex
11 LaTeX2e <2024-06-01> patch level 2
12 L3 programming layer <2024-05-27>
13 (./sn-jnl.cls
14 Document Class: sn-jnl 2019/11/18 v0.1: An authoring template for
15 Springer Jour
16 nal articles
17 (c:/texlive/2024/texmf-dist/tex/latex/base/article.cls
18 Document Class: article 2024/02/08 v1.4n Standard LaTeX document class
19 (c:/texlive/2024/texmf-dist/tex/latex/base/fleqn.clo
20 File: fleqn.clo 2016/12/29 v1.2b Standard LaTeX option (flush left
21 equations)
22 \mathindent=\skip49
23 Applying: [2015/01/01] Make \[ robust on input line 50.
24 LaTeX Info: Redefining \[ on input line 51.
25 Already applied: [0000/00/00] Make \[ robust on input line 62.
26 Applying: [2015/01/01] Make \] robust on input line 74.
27 LaTeX Info: Redefining \] on input line 75.
28 Already applied: [0000/00/00] Make \] robust on input line 83.
29 ) (c:/texlive/2024/texmf-dist/tex/latex/base/size10.clo
30 File: size10.clo 2024/02/08 v1.4n Standard LaTeX file (size option)
31 )
32 \c@part=\count194
33 \c@section=\count195
34 \c@subsection=\count196
35 \c@subsubsection=\count197
36 \c@paragraph=\count198
37 \c@subparagraph=\count199
38 \c@figure=\count266
39 \c@table=\count267
40 \abovecaptionskip=\skip50
41 \belowcaptionskip=\skip51
42 \bibindent=\dimen141
43 )
44 LaTeX Info: Redefining \rm on input line 146.
45 LaTeX Info: Redefining \sf on input line 147.
46 LaTeX Info: Redefining \tt on input line 148.
47 LaTeX Info: Redefining \bf on input line 149.
48 LaTeX Info: Redefining \it on input line 150.
49 LaTeX Info: Redefining \sl on input line 151.
50 LaTeX Info: Redefining \sc on input line 152.
51 LaTeX Info: Redefining \cal on input line 153.
52 LaTeX Info: Redefining \mit on input line 154.
53 LaTeX Info: Redefining \textsubscript on input line 212.
54 \columnhsize=\skip52
55 (c:/texlive/2024/texmf-dist/tex/latex/geometry/geometry.sty
56 Package: geometry 2020/01/02 v5.9 Page Geometry
57 (c:/texlive/2024/texmf-dist/tex/latex/graphics/keyval.sty
58
59
60
61
62
63
64
65
```

```

1
2
3
4 Package: keyval 2022/05/29 v1.15 key=value parser (DPC)
5 \KV@toks@=\toks17
6 ) (c:/texlive/2024/texmf-dist/tex/generic/iftex/ifvtex.sty
7 Package: ifvtex 2019/10/25 v1.7 ifvtex legacy package. Use iftex instead.
8 (c:/texlive/2024/texmf-dist/tex/generic/iftex/iftex.sty
9 Package: iftex 2022/02/03 v1.0f TeX engine tests
10 ))
11 \Gm@cnth=\count268
12 \Gm@cntv=\count269
13 \c@Gm@tempcnt=\count270
14 \Gm@bindingoffset=\dimen142
15 \Gm@wd@mp=\dimen143
16 \Gm@odd@mp=\dimen144
17 \Gm@even@mp=\dimen145
18 \Gm@layoutwidth=\dimen146
19 \Gm@layoutheight=\dimen147
20 \Gm@layouthoffset=\dimen148
21 \Gm@layoutvoffset=\dimen149
22 \Gm@dimlist=\toks18
23 )
24 \artcatbox=\box52
25 \auccount=\count271
26 \corraucount=\count272
27 \punctcount=\count273
28 \emailcnt=\count274
29 \c@affn=\count275
30 \addcount=\count276
31 \PacsCount=\count277
32 \PacsTmpCnt=\count278
33 \FMremarkdim=\dimen150
34 \fmremarkbox=\box53
35 (c:/texlive/2024/texmf-dist/tex/latex/sttools/cuted.sty
36 Package: cuted 2021/10/04 v2.0 Mixing onecolumn and twocolumn modes
37 \At@ViperColsBreak=\toks19
38 \preCutedStrip=\toks20
39 \postCutedStrip=\toks21
40 \cuted@@tempbox@a=\box54
41 \cuted@@tempbox@c=\box55
42 \cuted@@tempbox@var=\box56
43 \hold@viper=\box57
44 \@viper=\box58
45 \cuted@@varbox@a=\box59
46 \cuted@@varbox@c=\box60
47 \cuted@@tempdim@spread=\dimen151
48 \cuted@@tempdim@a=\dimen152
49 \cuted@@tempdim@b=\dimen153
50 \ht@hold@viper=\dimen154
51 \ht@viper=\dimen155
52 \var@@pagediscards@ht=\dimen156
53 \stripsep=\skip53
54 \viper@penalty=\count279
55 )
56 \firstpagehtcheck=\dimen157
57 \labelwidthi=\dimen158
58
59
60
61
62
63
64
65

```



```

1
2
3
4 \labelwidthiii=\dimen159
5 \labelwidthiiii=\dimen160
6 \labelwidthiv=\dimen161
7 \figwidth=\dimen162
8 \figheight=\dimen163
9 \sidecapwidth=\dimen164
10 \wrapcapline=\dimen165
11 \totalwrapline=\dimen166
12 \wraptotline=\dimen167
13 \figurebox=\box61
14 \wrapfigcapbox=\box62
15 \figcapbox=\box63
16 \capbox=\box64
17 \headwidthskip=\skip54
18 \tabcapbox=\box65
19 \temptbox=\box66
20 \tempdime=\dimen168
21 \tabhtdime=\dimen169
22
23 (c:/texlive/2024/texmf-dist/tex/latex/graphics/rotating.sty
24 Package: rotating 2016/08/11 v2.16d rotated objects in LaTeX
25 (c:/texlive/2024/texmf-dist/tex/latex/graphics/graphicx.sty
26 Package: graphicx 2021/09/16 v1.2d Enhanced LaTeX Graphics (DPC,SPQR)
27 (c:/texlive/2024/texmf-dist/tex/latex/graphics/graphics.sty
28 Package: graphics 2024/05/23 v1.4g Standard LaTeX Graphics (DPC,SPQR)
29 (c:/texlive/2024/texmf-dist/tex/latex/graphics/trig.sty
30 Package: trig 2023/12/02 v1.11 sin cos tan (DPC)
31 ) (c:/texlive/2024/texmf-dist/tex/latex/graphics-cfg/graphics.cfg
32 File: graphics.cfg 2016/06/04 v1.11 sample graphics configuration
33 )
34 Package graphics Info: Driver file: pdftex.def on input line 106.
35 (c:/texlive/2024/texmf-dist/tex/latex/graphics-def/pdftex.def
36 File: pdftex.def 2024/04/13 v1.2c Graphics/color driver for pdftex
37 ))
38 \Gin@req@height=\dimen170
39 \Gin@req@width=\dimen171
40 ) (c:/texlive/2024/texmf-dist/tex/latex/base/ifthen.sty
41 Package: ifthen 2024/03/16 v1.1e Standard LaTeX ifthen package (DPC)
42 )
43 \c@r@tfl@t=\count280
44 \rotFPtop=\skip55
45 \rotFPbot=\skip56
46 \rot@float@box=\box67
47 \rot@mess@toks=\toks22
48 ) (c:/texlive/2024/texmf-dist/tex/latex/threeparttable/threeparttable.sty
49 Package: threeparttable 2003/06/13 v 3.0
50 \@tempboxb=\box68
51 ) (c:/texlive/2024/texmf-dist/tex/latex/appendix/appendix.sty
52 Package: appendix 2020/02/08 v1.2c extra appendix facilities
53 \c@pps=\count281
54 \c@ppsavesec=\count282
55 \c@ppsaveapp=\count283
56 ) (c:/texlive/2024/texmf-dist/tex/latex/hyperref/hyperref.sty
57 Package: hyperref 2024-07-10 v7.01j Hypertext links for LaTeX
58 (c:/texlive/2024/texmf-dist/tex/latex/kvsetkeys/kvsetkeys.sty
59
60
61
62
63
64
65

```

```

1
2
3
4 Package: kvsetkeys 2022-10-05 v1.19 Key value parser (HO)
5 ) (c:/texlive/2024/texmf-dist/tex/generic/kvdefinekeys/kvdefinekeys.sty
6 Package: kvdefinekeys 2019-12-19 v1.6 Define keys (HO)
7 ) (c:/texlive/2024/texmf-dist/tex/generic/pdfescape/pdfescape.sty
8 Package: pdfescape 2019/12/09 v1.15 Implements pdfTeX's escape features
9 (HO)
10 (c:/texlive/2024/texmf-dist/tex/generic/ltxcmds/ltxcmds.sty
11 Package: ltxcmds 2023-12-04 v1.26 LaTeX kernel commands for general use
12 (HO)
13 ) (c:/texlive/2024/texmf-dist/tex/generic/pdftextcmds/pdftextcmds.sty
14 Package: pdftextcmds 2020-06-27 v0.33 Utility functions of pdfTeX for
15 LuaTeX (HO
16 )
17 (c:/texlive/2024/texmf-dist/tex/generic/infwarerr/infwarerr.sty
18 Package: infwarerr 2019/12/03 v1.5 Providing info/warning/error messages
19 (HO)
20 )
21 Package pdftextcmds Info: \pdf@primitive is available.
22 Package pdftextcmds Info: \pdf@ifprimitive is available.
23 Package pdftextcmds Info: \pdfdraftmode found.
24 )) (c:/texlive/2024/texmf-dist/tex/latex/hycolor/hycolor.sty
25 Package: hycolor 2020-01-27 v1.10 Color options for hyperref/bookmark
26 (HO)
27 ) (c:/texlive/2024/texmf-dist/tex/latex/hyperref/nameref.sty
28 Package: nameref 2023-11-26 v2.56 Cross-referencing by name of section
29 (c:/texlive/2024/texmf-dist/tex/latex/refcount/refcount.sty
30 Package: refcount 2019/12/15 v3.6 Data extraction from label references
31 (HO)
32 ) (c:/texlive/2024/texmf-
33 dist/tex/generic/gettitlestring/gettitlestring.sty
34 Package: gettitlestring 2019/12/15 v1.6 Cleanup title references (HO)
35 (c:/texlive/2024/texmf-dist/tex/latex/kvoptions/kvoptions.sty
36 Package: kvoptions 2022-06-15 v3.15 Key value format for package options
37 (HO)
38 ))
39 \c@section@level=\count284
40 ) (c:/texlive/2024/texmf-dist/tex/latex/etoolbox/etoolbox.sty
41 Package: etoolbox 2020/10/05 v2.5k e-TeX tools for LaTeX (JAW)
42 \etb@tempcnta=\count285
43 ) (c:/texlive/2024/texmf-dist/tex/generic/stringenc/stringenc.sty
44 Package: stringenc 2019/11/29 v1.12 Convert strings between diff.
45 encodings (HO
46 )
47 )
48 \@linkdim=\dimen172
49 \Hy@linkcounter=\count286
50 \Hy@pagecounter=\count287
51 (c:/texlive/2024/texmf-dist/tex/latex/hyperref/pd1enc.def
52 File: pd1enc.def 2024-07-10 v7.01j Hyperref: PDFDocEncoding definition
53 (HO)
54 Now handling font encoding PD1 ...
55 ... no UTF-8 mapping file for font encoding PD1
56 ) (c:/texlive/2024/texmf-dist/tex/generic/intcalc/intcalc.sty
57
58
59
60
61
62
63
64
65

```

```

1
2
3
4 Package: intcalc 2019/12/15 v1.3 Expandable calculations with integers
5 (HO)
6 )
7 \Hy@SavedSpaceFactor=\count288
8 (c:/texlive/2024/texmf-dist/tex/latex/hyperref/puenc.def
9 File: puenc.def 2024-07-10 v7.01j Hyperref: PDF Unicode definition (HO)
10 Now handling font encoding PU ...
11 ... no UTF-8 mapping file for font encoding PU
12 )
13 Package hyperref Info: Hyper figures OFF on input line 4157.
14 Package hyperref Info: Link nesting OFF on input line 4162.
15 Package hyperref Info: Hyper index ON on input line 4165.
16 Package hyperref Info: Plain pages OFF on input line 4172.
17 Package hyperref Info: Backreferencing OFF on input line 4177.
18 Package hyperref Info: Implicit mode ON; LaTeX internals redefined.
19 Package hyperref Info: Bookmarks ON on input line 4424.
20 \c@Hy@tempcnt=\count289
21 (c:/texlive/2024/texmf-dist/tex/latex/url/url.sty
22 \Urlmuskip=\muskip17
23 Package: url 2013/09/16 ver 3.4 Verb mode for urls, etc.
24 )
25 LaTeX Info: Redefining \url on input line 4763.
26 \XeTeXLinkMargin=\dimen173
27 (c:/texlive/2024/texmf-dist/tex/generic/bitset/bitset.sty
28 Package: bitset 2019/12/09 v1.3 Handle bit-vector datatype (HO)
29 (c:/texlive/2024/texmf-dist/tex/generic/bigintcalc/bigintcalc.sty
30 Package: bigintcalc 2019/12/15 v1.5 Expandable calculations on big
31 integers (HO
32 )
33 ))
34 \Fld@menulength=\count290
35 \Field@Width=\dimen174
36 \Fld@charsize=\dimen175
37 Package hyperref Info: Hyper figures OFF on input line 6042.
38 Package hyperref Info: Link nesting OFF on input line 6047.
39 Package hyperref Info: Hyper index ON on input line 6050.
40 Package hyperref Info: backreferencing OFF on input line 6057.
41 Package hyperref Info: Link coloring OFF on input line 6062.
42 Package hyperref Info: Link coloring with OCG OFF on input line 6067.
43 Package hyperref Info: PDF/A mode OFF on input line 6072.
44 (c:/texlive/2024/texmf-dist/tex/latex/base/atbegshi-ltx.sty
45 Package: atbegshi-ltx 2021/01/10 v1.0c Emulation of the original atbegshi
46 package with kernel methods
47 )
48 \Hy@abspage=\count291
49 \c@Item=\count292
50 \c@Hfootnote=\count293
51 )
52 Package hyperref Info: Driver (autodetected): hpdftex.
53 (c:/texlive/2024/texmf-dist/tex/latex/hyperref/hpdftex.def
54 File: hpdftex.def 2024-07-10 v7.01j Hyperref driver for pdfTeX
55 (c:/texlive/2024/texmf-dist/tex/latex/base/atveryend-ltx.sty
56 Package: atveryend-ltx 2020/08/19 v1.0a Emulation of the original
57 atveryend pac
58
59
60
61
62
63
64
65

```

```

kage
with kernel methods
)
\Fld@listcount=\count294
\c@bookmark@seq@number=\count295
(c:/texlive/2024/texmf-dist/tex/latex/rerunfilecheck/rerunfilecheck.sty
Package: rerunfilecheck 2022-07-10 v1.10 Rerun checks for auxiliary files
(HO)
(c:/texlive/2024/texmf-dist/tex/generic/uniquecounter/uniquecounter.sty
Package: uniquecounter 2019/12/15 v1.4 Provide unlimited unique counter
(HO)
)
Package uniquecounter Info: New unique counter `rerunfilecheck' on input
line 2
85.
)
\Hy@SectionHShift=\skip57
)
Package hyperref Info: Option `colorlinks' set `true' on input line 1471.
Package hyperref Info: Option `breaklinks' set `true' on input line 1471.
Package hyperref Info: Option `plainpages' set `false' on input line
1471.
Package hyperref Info: Option `bookmarksopen' set `true' on input line
1471.
Package hyperref Info: Option `bookmarksnumbered' set `false' on input
line 147
1.
(c:/texlive/2024/texmf-dist/tex/latex/wrapfig/wrapfig.sty
\wrapoverhang=\dimen176
\WF@size=\dimen177
\c@WF@wrappedlines=\count296
\WF@box=\box69
\WF@everypar=\toks23
Package: wrapfig 2003/01/31 v 3.6
)
\wraplines=\count297
\@authorfigbox=\box70
\@authorfigboxdim=\skip58
\biofigadjskip=\skip59
(c:/texlive/2024/texmf-dist/tex/latex/amscls/amsthm.sty
Package: amsthm 2020/05/29 v2.20.6
\thm@style=\toks24
\thm@bodyfont=\toks25
\thm@headfont=\toks26
\thm@notefont=\toks27
\thm@headpunct=\toks28
\thm@preskip=\skip60
\thm@postskip=\skip61
\thm@headsep=\skip62
\dth@everypar=\toks29
) (c:/texlive/2024/texmf-dist/tex/latex/base/fix-cm.sty
Package: fix-cm 2020/11/24 v1.1t fixes to LaTeX
(c:/texlive/2024/texmf-dist/tex/latex/base/tslenc.def
File: tslenc.def 2001/06/05 v3.0e (jk/car/fm) Standard LaTeX file

```

```

LaTeX Font Info:   Redeclaring font encoding TS1 on input line 47.
)) (c:/texlive/2024/texmf-dist/tex/latex/natbib/natbib.sty
Package: natbib 2010/09/13 8.31b (PWD, AO)
\bibhang=\skip63
\bibsep=\skip64
LaTeX Info: Redefining \cite on input line 694.
\c@NAT@ctr=\count298
)) (c:/texlive/2024/texmf-dist/tex/latex/multirow/multirow.sty
Package: multirow 2021/03/15 v2.8 Span multiple rows of a table
\multirow@colwidth=\skip65
\multirow@cntb=\count299
\multirow@dima=\skip66
\bigstrutjot=\dimen178
) (c:/texlive/2024/texmf-dist/tex/latex/amsmath/amsmath.sty
Package: amsmath 2024/05/23 v2.17q AMS math features
\@mathmargin=\skip67
For additional information on amsmath, use the '?' option.
(c:/texlive/2024/texmf-dist/tex/latex/amsmath/amstext.sty
Package: amstext 2021/08/26 v2.01 AMS text
(c:/texlive/2024/texmf-dist/tex/latex/amsmath/amsgen.sty
File: amsgen.sty 1999/11/30 v2.0 generic functions
\@emptytoks=\toks30
\ex@=\dimen179
)) (c:/texlive/2024/texmf-dist/tex/latex/amsmath/amsbsy.sty
Package: amsbsy 1999/11/29 v1.2d Bold Symbols
\pmbraise@=\dimen180
) (c:/texlive/2024/texmf-dist/tex/latex/amsmath/amsopn.sty
Package: amsopn 2022/04/08 v2.04 operator names
)
\inf@bad=\count300
LaTeX Info: Redefining \frac on input line 233.
\uproot@=\count301
\leftroot@=\count302
LaTeX Info: Redefining \overline on input line 398.
LaTeX Info: Redefining \colon on input line 409.
\classnum@=\count303
\DOTSCASE@=\count304
LaTeX Info: Redefining \ldots on input line 495.
LaTeX Info: Redefining \dots on input line 498.
LaTeX Info: Redefining \cdots on input line 619.
\Mathstrutbox@=\box71
\strutbox@=\box72
LaTeX Info: Redefining \big on input line 721.
LaTeX Info: Redefining \Big on input line 722.
LaTeX Info: Redefining \bigg on input line 723.
LaTeX Info: Redefining \Bigg on input line 724.
\big@size=\dimen181
LaTeX Font Info:   Redeclaring font encoding OML on input line 742.
LaTeX Font Info:   Redeclaring font encoding OMS on input line 743.
\maccc@depth=\count305
LaTeX Info: Redefining \bmod on input line 904.
LaTeX Info: Redefining \pmod on input line 909.
LaTeX Info: Redefining \smash on input line 939.
LaTeX Info: Redefining \relbar on input line 969.

```

```

LaTeX Info: Redefining \Relbar on input line 970.
\c@MaxMatrixCols=\count306
\dotsspace@=\muskip18
\c@parentequation=\count307
\dspbrk@lvl=\count308
\tag@help=\toks31
\row@=\count309
\column@=\count310
\maxfields@=\count311
\andhelp@=\toks32
\eqnshift@=\dimen182
\alignsep@=\dimen183
\tagshift@=\dimen184
\tagwidth@=\dimen185
\totwidth@=\dimen186
\lineht@=\dimen187
\@envbody=\toks33
\multlinegap=\skip68
\multlinetaggap=\skip69
\mathdisplay@stack=\toks34
LaTeX Info: Redefining \[ on input line 2953.
LaTeX Info: Redefining \] on input line 2954.
) (c:/texlive/2024/texmf-dist/tex/latex/amsfonts/amssymb.sty
Package: amssymb 2013/01/14 v3.01 AMS font symbols
(c:/texlive/2024/texmf-dist/tex/latex/amsfonts/amsfonts.sty
Package: amsfonts 2013/01/14 v3.01 Basic AMSFonts support
\symAMSa=\mathgroup4
\symAMSb=\mathgroup5
LaTeX Font Info: Redefining math symbol \hbar on input line 98.
LaTeX Font Info: Overwriting math alphabet '\mathfrak' in version
'bold'
(Font) U/euf/m/n --> U/euf/b/n on input line 106.
)) (c:/texlive/2024/texmf-dist/tex/latex/jknapltx/mathrsfs.sty
Package: mathrsfs 1996/01/01 Math RSFS package v1.0 (jk)
\symrsfs=\mathgroup6
) (c:/texlive/2024/texmf-dist/tex/latex/xcolor/xcolor.sty
Package: xcolor 2023/11/15 v3.01 LaTeX color extensions (UK)
(c:/texlive/2024/texmf-dist/tex/latex/graphics-cfg/color.cfg
File: color.cfg 2016/01/02 v1.6 sample color configuration
)
Package xcolor Info: Driver file: pdftex.def on input line 274.
(c:/texlive/2024/texmf-dist/tex/latex/graphics/mathcolor.ltx)
Package xcolor Info: Model 'cmy' substituted by 'cmy0' on input line
1350.
Package xcolor Info: Model 'hsb' substituted by 'rgb' on input line 1354.
Package xcolor Info: Model 'RGB' extended on input line 1366.
Package xcolor Info: Model 'HTML' substituted by 'rgb' on input line
1368.
Package xcolor Info: Model 'Hsb' substituted by 'hsb' on input line 1369.
Package xcolor Info: Model 'tHsb' substituted by 'hsb' on input line
1370.
Package xcolor Info: Model 'HSB' substituted by 'hsb' on input line 1371.
Package xcolor Info: Model 'Gray' substituted by 'gray' on input line
1372.

```

```

Package xcolor Info: Model `wave' substituted by `hsb' on input line
1373.
) (c:/texlive/2024/texmf-dist/tex/latex/base/textcomp.sty
Package: textcomp 2024/04/24 v2.1b Standard LaTeX package
) (c:/texlive/2024/texmf-dist/tex/latex/ncctools/manyfoot.sty
Package: manyfoot 2019/08/03 v1.11 Many Footnote Levels Package (NCC)
(c:/texlive/2024/texmf-dist/tex/latex/ncctools/nccfoots.sty
Package: nccfoots 2005/02/03 v1.2 NCC Footnotes Package (NCC)
)
\MFL@columnwidth=\dimen188
) (c:/texlive/2024/texmf-dist/tex/latex/booktabs/booktabs.sty
Package: booktabs 2020/01/12 v1.61803398 Publication quality tables
\heavyrulewidth=\dimen189
\lightrulewidth=\dimen190
\cmidrulewidth=\dimen191
\belowrulesep=\dimen192
\belowbottomsep=\dimen193
\aboverulesep=\dimen194
\abovetopsep=\dimen195
\cmidrulesep=\dimen196
\cmidrulekern=\dimen197
\defaultaddspace=\dimen198
\@cmidla=\count312
\@cmidlb=\count313
\@aboverulesep=\dimen199
\@belowrulesep=\dimen256
\@thisruleclass=\count314
\@lastruleclass=\count315
\@thisrulewidth=\dimen257
) (c:/texlive/2024/texmf-dist/tex/latex/algorithms/algorithm.sty
Package: algorithm 2009/08/24 v0.1 Document Style `algorithm' - floating
enviro
nment
(c:/texlive/2024/texmf-dist/tex/latex/float/float.sty
Package: float 2001/11/08 v1.3d Float enhancements (AL)
\c@float@type=\count316
\float@exts=\toks35
\float@box=\box73
\@float@everytoks=\toks36
\@floatcapt=\box74
)
\@float@every@algorithm=\toks37
\c@algorithm=\count317
) (c:/texlive/2024/texmf-dist/tex/latex/algorithmicx/algorithmicx.sty
Package: algorithmicx 2005/04/27 v1.2 Algorithmicx
Document Style algorithmicx 1.2 - a greatly improved `algorithmic' style
\c@ALG@line=\count318
\c@ALG@rem=\count319
\c@ALG@nested=\count320
\ALG@tlm=\skip70
\ALG@thistlm=\skip71
\c@ALG@Lnr=\count321
\c@ALG@blocknr=\count322
\c@ALG@storecount=\count323

```

```

1
2
3
4 \c@ALG@tmpcounter=\count324
5 \ALG@tmplength=\skip72
6 ) (c:/texlive/2024/texmf-dist/tex/latex/algorithmicx/algpseudocode.sty
7 Package: algpseudocode
8 Document Style - pseudocode environments for use with the `algorithmicx'
9 style
10 ) (c:/texlive/2024/texmf-dist/tex/latex/listings/listings.sty
11 \lst@mode=\count325
12 \lst@gtempboxa=\box75
13 \lst@token=\toks38
14 \lst@length=\count326
15 \lst@currlwidth=\dimen258
16 \lst@column=\count327
17 \lst@pos=\count328
18 \lst@lostspace=\dimen259
19 \lst@width=\dimen260
20 \lst@newlines=\count329
21 \lst@lineno=\count330
22 \lst@maxwidth=\dimen261
23 (c:/texlive/2024/texmf-dist/tex/latex/listings/lstpatch.sty
24 File: lstpatch.sty 2024/05/25 1.10b (Carsten Heinz)
25 ) (c:/texlive/2024/texmf-dist/tex/latex/listings/lstmisc.sty
26 File: lstmisc.sty 2024/05/25 1.10b (Carsten Heinz)
27 \c@lstnumber=\count331
28 \lst@skipnumbers=\count332
29 \lst@framebox=\box76
30 ) (c:/texlive/2024/texmf-dist/tex/latex/listings/listings.cfg
31 File: listings.cfg 2024/05/25 1.10b listings configuration
32 ))
33 Package: listings 2024/05/25 1.10b (Carsten Heinz)
34 \c@theorem=\count333
35 \c@example=\count334
36 \c@remark=\count335
37 \c@definition=\count336
38 (./setting.tex (c:/texlive/2024/texmf-
39 dist/tex/latex/arydshln/arydshln.sty
40 Package: arydshln 2019/02/21 v1.76
41 \dashlinedash=\dimen262
42 \dashlinegap=\dimen263
43 \adl@box=\box77
44 \adl@height=\dimen264
45 \adl@heightsave=\dimen265
46 \adl@depth=\dimen266
47 \adl@depthsave=\dimen267
48 \adl@finaldepth=\dimen268
49 \adl@columns=\count337
50 \adl@ncol=\count338
51 \adl@currentcolumn=\count339
52 \adl@currentcolumnsave=\count340
53 \adl@totalheight=\count341
54 \adl@totalheightsave=\count342
55 \adl@dash=\count343
56 \adl@gap=\count344
57 \adl@cla=\count345
58
59
60
61
62
63
64
65

```



```

\adl@clb=\count346
\adl@everyvbox=\toks39
\adl@LTpagetotal=\dimen269
) (c:/texlive/2024/texmf-dist/tex/latex/mathtools/mathtools.sty
Package: mathtools 2024/03/11 v1.30 mathematical typesetting tools
(c:/texlive/2024/texmf-dist/tex/latex/tools/calc.sty
Package: calc 2023/07/08 v4.3 Infix arithmetic (KKT,FJ)
\calc@Acount=\count347
\calc@Bcount=\count348
\calc@Adimen=\dimen270
\calc@Bdimen=\dimen271
\calc@Askip=\skip73
\calc@Bskip=\skip74
LaTeX Info: Redefining \setlength on input line 80.
LaTeX Info: Redefining \addtolength on input line 81.
\calc@Ccount=\count349
\calc@Cskip=\skip75
) (c:/texlive/2024/texmf-dist/tex/latex/mathtools/mhsetup.sty
Package: mhsetup 2021/03/18 v1.4 programming setup (MH)
)
\g_MT_multlinerow_int=\count350
\l_MT_multwidth_dim=\dimen272
\origjot=\skip76
\l_MT_shortvdotswithinadjustabove_dim=\dimen273
\l_MT_shortvdotswithinadjustbelow_dim=\dimen274
\l_MT_above_intertext_sep=\dimen275
\l_MT_below_intertext_sep=\dimen276
\l_MT_above_shortintertext_sep=\dimen277
\l_MT_below_shortintertext_sep=\dimen278
\xmathstrut@box=\box78
\xmathstrut@dim=\dimen279
) (c:/texlive/2024/texmf-dist/tex/latex/doublestroke/dsfont.sty
Package: dsfont 1995/08/01 v0.1 Double stroke roman fonts
) (c:/texlive/2024/texmf-dist/tex/latex/circledsteps/circledsteps.sty
Package: circledsteps 2023/08/11 v1.3.3 Circled steps that you can
reference
(c:/texlive/2024/texmf-dist/tex/latex/pict2e/pict2e.sty
Package: pict2e 2020/09/30 v0.4b Improved picture commands (HjG,RN,JT)
(c:/texlive/2024/texmf-dist/tex/latex/pict2e/pict2e.cfg
File: pict2e.cfg 2016/02/05 v0.1u pict2e configuration for teTeX/TeXLive
)
Package pict2e Info: Driver file: pdftex.def on input line 112.
Package pict2e Info: Driver file for pict2e: p2e-pdftex.def on input line
114.
(c:/texlive/2024/texmf-dist/tex/latex/pict2e/p2e-pdftex.def
File: p2e-pdftex.def 2016/02/05 v0.1u Driver-dependant file (RN,HjG,JT)
)
\pIIe@GRAPH=\toks40
\@arclen=\dimen280
\@arcrad=\dimen281
\pIIe@tempdima=\dimen282
\pIIe@tempdimb=\dimen283
\pIIe@tempdimc=\dimen284
\pIIe@tempdimd=\dimen285

```

```

\pIIe@tempdime=\dimen286
\pIIe@tempdimf=\dimen287
) (c:/texlive/2024/texmf-dist/tex/latex/picture/picture.sty
Package: picture 2020-04-22 v1.6 Dimens for picture macros (HO)
Package picture Info: Calculation method: calc.
) (c:/texlive/2024/texmf-dist/tex/latex/pgf/utilities/pgfkeys.sty
(c:/texlive/2
024/texmf-dist/tex/generic/pgf/utilities/pgfkeys.code.tex
\pgfkeys@pathtoks=\toks41
\pgfkeys@temptoks=\toks42

(c:/texlive/2024/texmf-
dist/tex/generic/pgf/utilities/pgfkeyslibraryfiltered.co
de.tex
\pgfkeys@tmptoks=\toks43
)))
\csteps@CBox=\box79
\csteps@XLength=\skip77
\csteps@YLength=\skip78
\csteps@YDepth=\skip79
\csteps@tmplen=\skip80
\c@stepcnt=\count351
) (c:/texlive/2024/texmf-dist/tex/latex/caption/subcaption.sty
Package: subcaption 2023/07/28 v1.6b Sub-captions (AR)
(c:/texlive/2024/texmf-dist/tex/latex/caption/caption.sty
Package: caption 2023/08/05 v3.6o Customizing captions (AR)
(c:/texlive/2024/texmf-dist/tex/latex/caption/caption3.sty
Package: caption3 2023/07/31 v2.4d caption3 kernel (AR)
\caption@tempdima=\dimen288
\captionmargin=\dimen289
\caption@leftmargin=\dimen290
\caption@rightmargin=\dimen291
\caption@width=\dimen292
\caption@indent=\dimen293
\caption@parindent=\dimen294
\caption@hangindent=\dimen295
Package caption Info: Unknown document class (or package),
(caption) standard defaults will be used.
Package caption Info: \@makecaption = \long macro:#1#2->\ifx \FigName
\@capttype
\vskip \abovecaptionskip \@figurecaption {#1}{#2} \else \@tablecaption
{#1}{#2}
} \vskip \belowcaptionskip \fi on input line 1175.
)

Package caption Warning: Unknown document class (or package),
(caption) standard defaults will be used.
See the caption package documentation for explanation.

\c@caption@flags=\count352
\c@continuedfloat=\count353
Package caption Info: float package is loaded.
Package caption Info: hyperref package is loaded.
Package caption Info: listings package is loaded.

```

```

1
2
3
4 Package caption Info: rotating package is loaded.
5 Package caption Info: threeparttable package is loaded.
6 Package caption Info: wrapfig package is loaded.
7 )
8 Package caption Info: New subtype `subfigure' on input line 238.
9 \c@subfigure=\count354
10 Package caption Info: New subtype `subtable' on input line 238.
11 \c@subtable=\count355
12 )) (c:/texlive/2024/texmf-dist/tex/latex/l3backend/l3backend-pdfTeX.def
13 File: l3backend-pdfTeX.def 2024-05-08 L3 backend support: PDF output
14 (pdfTeX)
15 \l__color_backend_stack_int=\count356
16 \l__pdf_internal_box=\box80
17 ) (./root.aux)
18 \openout1 = `root.aux'.
19
20
21 LaTeX Font Info: Checking defaults for OML/cmm/m/it on input line 42.
22 LaTeX Font Info: ... okay on input line 42.
23 LaTeX Font Info: Checking defaults for OMS/cmsy/m/n on input line 42.
24 LaTeX Font Info: ... okay on input line 42.
25 LaTeX Font Info: Checking defaults for OT1/cmr/m/n on input line 42.
26 LaTeX Font Info: ... okay on input line 42.
27 LaTeX Font Info: Checking defaults for T1/cmr/m/n on input line 42.
28 LaTeX Font Info: ... okay on input line 42.
29 LaTeX Font Info: Checking defaults for TS1/cmr/m/n on input line 42.
30 LaTeX Font Info: ... okay on input line 42.
31 LaTeX Font Info: Checking defaults for OMX/cmex/m/n on input line 42.
32 LaTeX Font Info: ... okay on input line 42.
33 LaTeX Font Info: Checking defaults for U/cmr/m/n on input line 42.
34 LaTeX Font Info: ... okay on input line 42.
35 LaTeX Font Info: Checking defaults for PD1/pdf/m/n on input line 42.
36 LaTeX Font Info: ... okay on input line 42.
37 LaTeX Font Info: Checking defaults for PU/pdf/m/n on input line 42.
38 LaTeX Font Info: ... okay on input line 42.
39
40 *geometry* driver: auto-detecting
41 *geometry* detected driver: pdftex
42 *geometry* verbose mode - [ preamble ] result:
43 * driver: pdftex
44 * paper: custom
45 * layout: <same size as paper>
46 * layoutoffset: (h,v)=(0.0pt,0.0pt)
47 * bindingoffset: 17.07164pt
48 * modes: twoside
49 * h-part: (L,W,R)=(83.37448pt, 372.0pt, 125.06175pt)
50 * v-part: (T,H,B)=(73.97716pt, 552.69478pt, 218.37491pt)
51 * \paperwidth=597.50787pt
52 * \paperheight=845.04684pt
53 * \textwidth=372.0pt
54 * \textheight=552.69478pt
55 * \oddsidemargin=28.17613pt
56 * \evensidemargin=52.79176pt
57 * \topmargin=-19.72638pt
58 * \headheight=5.5pt
59 * \headsep=15.93355pt
60
61
62
63
64
65

```

```

1
2
3
4 * \topskip=10.0pt
5 * \footskip=28.45274pt
6 * \marginparwidth=34.1433pt
7 * \marginparsep=14.22636pt
8 * \columnsep=14.22636pt
9 * \skip\footins=8.0pt plus 4.0pt
10 * \hoffset=0.0pt
11 * \voffset=0.0pt
12 * \mag=1000
13 * \@twocolumnfalse
14 * \@twoside true
15 * \mparswitch true
16 * \reversemargin true
17 * (lin=72.27pt=25.4mm, 1cm=28.453pt)
18
19
20 (c:/texlive/2024/texmf-dist/tex/context/base/mkii/supp-pdf.mkii
21 [Loading MPS to PDF converter (version 2006.09.02).]
22 \scratchcounter=\count357
23 \scratchdimen=\dimen296
24 \scratchbox=\box81
25 \nofMPsegments=\count358
26 \nofMParguments=\count359
27 \everyMPshowfont=\toks44
28 \MPscratchCnt=\count360
29 \MPscratchDim=\dimen297
30 \MPnumerator=\count361
31 \makeMPintoPDFobject=\count362
32 \everyMPtoPDFconversion=\toks45
33 ) (c:/texlive/2024/texmf-dist/tex/latex/epstopdf-pkg/epstopdf-base.sty
34 Package: epstopdf-base 2020-01-24 v2.11 Base part for package epstopdf
35 Package epstopdf-base Info: Redefining graphics rule for '.eps' on input
36 line 4
37 85.
38 (c:/texlive/2024/texmf-dist/tex/latex/latexconfig/epstopdf-sys.cfg
39 File: epstopdf-sys.cfg 2010/07/13 v1.3 Configuration of (r)epstopdf for
40 TeX Liv
41 e
42 ))
43 LaTeX Info: Redefining \S on input line 42.
44 LaTeX Font Info: Redefining symbol font 'AMSA' on input line 42.
45 LaTeX Font Info: Overwriting symbol font 'AMSA' in version 'normal'
46 (Font) U/msa/m/n --> U/msa/m/n on input line 42.
47 LaTeX Font Info: Overwriting symbol font 'AMSA' in version 'bold'
48 (Font) U/msa/m/n --> U/msa/m/n on input line 42.
49 Package hyperref Info: Link coloring ON on input line 42.
50 (./root.out) (./root.out)
51 \outlinefile=\write3
52 \openout3 = `root.out'.
53
54
55 \c@lstlisting=\count363
56 Package caption Info: Begin \AtBeginDocument code.
57 Package caption Info: End \AtBeginDocument code.
58 LaTeX Font Info: Calculating math sizes for size <12.045> on input
59 line 86.
60
61
62
63
64
65

```

LaTeX Font Info: Trying to load font information for U+msa on input line 86.

(c:/texlive/2024/texmf-dist/tex/latex/amsfonts/umsa.fd
File: umsa.fd 2013/01/14 v3.01 AMS symbols A
)

LaTeX Font Info: Trying to load font information for U+msb on input line 86.

(c:/texlive/2024/texmf-dist/tex/latex/amsfonts/umsb.fd
File: umsb.fd 2013/01/14 v3.01 AMS symbols B
)

LaTeX Font Info: Trying to load font information for U+rsfs on input line 86

.
(c:/texlive/2024/texmf-dist/tex/latex/jknaptx/ursfs.fd
File: ursfs.fd 1998/03/24 rsfs font definition file (jk)
)

LaTeX Font Warning: Font shape `U/rsfs/m/n' in size <8.43146> not available
(Font) size <8> substituted on input line 86.

LaTeX Font Info: Calculating math sizes for size <11.04124> on input line 86

.

LaTeX Font Warning: Font shape `U/rsfs/m/n' in size <5.52061> not available
(Font) size <6> substituted on input line 86.

Underfull \vbox (badness 10000) has occurred while \output is active []
[1

{c:/texlive/2024/texmf-var/fonts/map/pdftex/updmap/pdftex.map}]
LaTeX Font Info: Calculating math sizes for size <10.03749> on input line 86

.

LaTeX Warning: File `figures/eyecatch_v3.pdf' not found on input line 94.

! Package pdftex.def Error: File `figures/eyecatch_v3.pdf' not found:
using dra
ft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

1.94 ...th=0.6\linewidth]{figures/eyecatch_v3.pdf}

Try typing <return> to proceed.

If that doesn't work, type X <return> to quit.

Package natbib Warning: Citation `KANTADAS2024221' on page 2 undefined on input line 104.

Package natbib Warning: Citation `UEDA2024139928' on page 2 undefined on input line 104.

Package natbib Warning: Citation `Kiyokawa2025dis' on page 2 undefined on input line 106.

Package natbib Warning: Citation `9419740' on page 2 undefined on input line 106.

Package natbib Warning: Citation `9492060' on page 2 undefined on input line 106.

Package natbib Warning: Citation `8653968' on page 2 undefined on input line 106.

Package natbib Warning: Citation `9966530' on page 2 undefined on input line 110.

Underfull \vbox (badness 10000) has occurred while \output is active []

[2]

Package natbib Warning: Citation `qi2017pointnet' on page 3 undefined on input line 120.

Package natbib Warning: Citation `wang2019dynamic' on page 3 undefined on input

line 120.

Package natbib Warning: Citation `hanocka2019meshcnn' on page 3 undefined
on input line 120.

Package natbib Warning: Citation `NIPS2015_8d55a249' on page 3 undefined
on input line 121.

Package natbib Warning: Citation `mirza2014conditional' on page 3
undefined on input line 121.

Package natbib Warning: Citation `sohl2015deep' on page 3 undefined on
input line 125.

Package natbib Warning: Citation `ho2020denoising' on page 3 undefined on
input line 125.

Package natbib Warning: Citation `dhariwal2021diffusion' on page 3
undefined on input line 125.

Package natbib Warning: Citation `lugmayr2022repaint' on page 3 undefined
on input line 125.

Overfull \vbox (2.40366pt too high) has occurred while \output is active
[]

Underfull \vbox (badness 10000) has occurred while \output is active []

[3{c:/texlive/2024/texmf-dist/fonts/enc/dvips/cm-super/cm-super-tsl.enc}]

Package natbib Warning: Citation `MISHRA2024100900' on page 4 undefined
on input line 151.

Package natbib Warning: Citation `9966530' on page 4 undefined on input
line 15
1.

Package natbib Warning: Citation `10155458' on page 4 undefined on input
line 1
51.

Package natbib Warning: Citation `Kiyokawa2025dis' on page 4 undefined on
input
line 152.

Package natbib Warning: Citation `9419740' on page 4 undefined on input
line 15
2.

Package natbib Warning: Citation `9492060' on page 4 undefined on input
line 15
2.

Package natbib Warning: Citation `8653968' on page 4 undefined on input
line 15
2.

Package natbib Warning: Citation `LU20221' on page 4 undefined on input
line 15
6.

Package natbib Warning: Citation `9551431' on page 4 undefined on input
line 15
6.

Package natbib Warning: Citation `TSUNAZAWA2018474' on page 4 undefined
on input
line 157.

Package natbib Warning: Citation `LI201988' on page 4 undefined on input
line 1
60.

Package natbib Warning: Citation `STERKENS2021105246' on page 4 undefined
on input
line 160.

Package natbib Warning: Citation `UEDA2024139928' on page 4 undefined on
input
line 161.

Package natbib Warning: Citation `sohl2015deep' on page 4 undefined on
input li
ne 168.

Package natbib Warning: Citation `ho2020denoising' on page 4 undefined on
input
line 168.

Package natbib Warning: Citation `dhariwal2021diffusion' on page 4
undefined on
input line 168.

Package natbib Warning: Citation `ramesh2022hierarchical' on page 4
undefined o
n input line 169.

Package natbib Warning: Citation `saharia2022photorealistic' on page 4
undefine
d on input line 169.

Package natbib Warning: Citation `zhao2023uni' on page 4 undefined on
input lin
e 169.

Package natbib Warning: Citation `pmlr-v235-cohen24a' on page 4 undefined
on in
put line 169.

Package natbib Warning: Citation `Kara_2024_CVPR' on page 4 undefined on
input
line 169.

Package natbib Warning: Citation `zhang2023towards' on page 4 undefined
on inpu
t line 169.

Package natbib Warning: Citation `janner2022diffuser' on page 4 undefined
on in
put line 169.

1
2
3
4
5
6 Package natbib Warning: Citation `mishra2023generative' on page 4
7 undefined on
8 input line 169.
9

10
11 Package natbib Warning: Citation `Ze2024DP3' on page 4 undefined on input
12 line
13 169.
14
15

16
17 Package natbib Warning: Citation `9578791' on page 4 undefined on input
18 line 17
19 1.
20
21

22 Package natbib Warning: Citation `NEURIPS2022_40e56dab' on page 4
23 undefined on
24 input line 172.
25
26

27 Package natbib Warning: Citation `nichol2022point' on page 4 undefined on
28 input
29 line 173.
30
31

32 Package natbib Warning: Citation `10656999' on page 4 undefined on input
33 line 1
34 74.
35
36

37 Package natbib Warning: Citation `liu2023meshdiffusion' on page 4
38 undefined on
39 input line 174.
40
41

42 Package natbib Warning: Citation `10658125' on page 4 undefined on input
43 line 1
44 74.
45
46

47 Package natbib Warning: Citation `wynn2023diffusionerf' on page 4
48 undefined on
49 input line 174.
50
51

52 Package natbib Warning: Citation `qi2017pointnet' on page 4 undefined on
53 input
54 line 176.
55
56

57 Package natbib Warning: Citation `wang2019dynamic' on page 4 undefined on
58 input
59
60
61
62
63
64
65

line 176.

Package natbib Warning: Citation `hanocka2019meshcnn' on page 4 undefined on input line 176.

Underfull \vbox (badness 10000) has occurred while \output is active [4]

Package natbib Warning: Citation `ho2021classifierfree' on page 5 undefined on input line 195.

Package natbib Warning: Citation `10203721' on page 5 undefined on input line 95.

Package natbib Warning: Citation `sohl2015deep' on page 5 undefined on input line 197.

Package natbib Warning: Citation `ho2020denoising' on page 5 undefined on input line 197.

Package natbib Warning: Citation `dhariwal2021diffusion' on page 5 undefined on input line 197.

Package natbib Warning: Citation `ho2020denoising' on page 5 undefined on input line 209.

Underfull \vbox (badness 10000) has occurred while \output is active [5]

Package natbib Warning: Citation `rombach2022high' on page 6 undefined on input line 218.

Package natbib Warning: Citation `ho2021classifierfree' on page 6
undefined on
input line 218.

Package natbib Warning: Citation `dhariwal2021diffusion' on page 6
undefined on
input line 224.

Package natbib Warning: Citation `ho2021classifierfree' on page 6
undefined on
input line 227.

LaTeX Warning: File `figures/overview_v1.pdf' not found on input line
252.

! Package pdftex.def Error: File `figures/overview_v1.pdf' not found:
using dra
ft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.

...

1.252 ...=1.0 \linewidth]{figures/overview_v1.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

Underfull \vbox (badness 10000) has occurred while \output is active []
[6]

Package natbib Warning: Citation `11127844' on page 7 undefined on input
line 2
72.

Package natbib Warning: Citation `11127730' on page 7 undefined on input
line 2
72.

Package natbib Warning: Citation `NEURIPS2023_4c5722ba' on page 7
undefined on
input line 272.

Underfull \vbox (badness 10000) has occurred while \output is active []
[7]

LaTeX Warning: File `figures/shape_to_voxel_convet_v2.pdf' not found on
input line 294.

! Package pdftex.def Error: File `figures/shape_to_voxel_convet_v2.pdf'
not found:
using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

l.294 ...th]{figures/shape_to_voxel_convet_v2.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

LaTeX Warning: File `figures/cutting_plan_v0.pdf' not found on input line
339.

! Package pdftex.def Error: File `figures/cutting_plan_v0.pdf' not found:
using
draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

l.339 ...columnwidth]{figures/cutting_plan_v0.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

Underfull \vbox (badness 10000) has occurred while \output is active []
[8]

Underfull \vbox (badness 10000) has occurred while \output is active []
[9]

Underfull \vbox (badness 10000) has occurred while \output is active []
[10]

Package natbib Warning: Citation `RiskMetrics1996' on page 11 undefined on input line 374.

Package natbib Warning: Citation `Stan_2000_CVaR' on page 11 undefined on input line 374.

LaTeX Font Info: Trying to load font information for U+dsrom on input line 377.
(c:/texlive/2024/texmf-dist/tex/latex/doublestroke/Udsrom.fd
File: Udsrom.fd 1995/08/01 v0.1 Double stroke roman font definitions
)

Package natbib Warning: Citation `Kaiming_2017_maskrcnn' on page 11 undefined on input line 411.

Package natbib Warning: Citation `Kirillov_2023_ICCV' on page 11 undefined on input line 411.

Package natbib Warning: Citation `Ravi_2025_sam2' on page 11 undefined on input line 411.

Package natbib Warning: Citation `carion_2026_sam3' on page 11 undefined on input line 411.

Underfull \vbox (badness 10000) has occurred while \output is active []
[11]

Package natbib Warning: Citation `sullivan2019pyvista' on page 12 undefined on input line 413.

Package natbib Warning: Citation `blender' on page 12 undefined on input line 4
13.

LaTeX Warning: File `figures/benchmark_shape2.pdf' not found on input line 422.

! Package pdftex.def Error: File `figures/benchmark_shape2.pdf' not found: using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

l.422 ...\\linewidth]{figures/benchmark_shape2.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

Underfull \vbox (badness 10000) has occurred while \output is active []
[12]

Package natbib Warning: Citation `ivanov2018variational' on page 13 undefined on input line 436.

Package natbib Warning: Citation `bishop2006prml' on page 13 undefined on input line 468.

Underfull \vbox (badness 10000) has occurred while \output is active []
[13]

Package natbib Warning: Citation `ivanov2018variational' on page 14 undefined on input line 471.

Package natbib Warning: Citation `Pathak2016ContextEF' on page 14 undefined on input line 471.

Package natbib Warning: Citation `song_2021_ddim' on page 14 undefined on input line 473.

LaTeX Warning: File `figures/sim_result/simple_model/simple_model6_cutting_process2.pdf' not found on input line 481.

! Package pdftex.def Error: File `figures/sim_result/simple_model/simple_model6_cutting_process2.pdf' not found: using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

l.481 ...model/simple_model6_cutting_process2.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

LaTeX Warning: File `figures/sim_result/simple_model/simple_model6_render_comp_2.pdf' not found on input line 596.

! Package pdftex.def Error: File `figures/sim_result/simple_model/simple_model6_render_comp_2.pdf' not found: using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

l.596 ...le_model/simple_model6_render_comp_2.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

LaTeX Warning: File `figures/sim_result/simple_model/simple_model8_render_comp_2.pdf' not found on input line 604.

! Package pdftex.def Error: File `figures/sim_result/simple_model/simple_model8_render_comp_2.pdf' not found: using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

1.604 ...le_model/simple_model8_render_comp_2.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

LaTeX Warning: File
'figures/sim_result/simple_model/simple_model1_render_comp_2.pdf' not found on input line 612.

! Package pdftex.def Error: File
'figures/sim_result/simple_model/simple_model1_render_comp_2.pdf' not found: using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

1.612 ...le_model/simple_model1_render_comp_2.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

LaTeX Warning: Float too large for page by 508.04878pt on input line 620.

Underfull \vbox (badness 10000) has occurred while \output is active []
[14]

Underfull \vbox (badness 10000) has occurred while \output is active []
[15]

Underfull \vbox (badness 10000) has occurred while \output is active []
[16]

Underfull \vbox (badness 10000) has occurred while \output is active []
[17]

Package natbib Warning: Citation `ivanov2018variational' on page 18
undefined on
input line 646.

Package natbib Warning: Citation `Pathak2016ContextEF' on page 18
undefined on
input line 646.

Package natbib Warning: Citation `song_2021_ddim' on page 18 undefined on
input
line 649.

LaTeX Warning: File
`figures/sim_result/complex_model/complex_model_cutting_sequences_v00.pdf' not found on input line 666.

! Package pdftex.def Error: File
`figures/sim_result/complex_model/complex_model_cutting_sequences_v00.pdf' not found: using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.

...
1.666 .../complex_model_cutting_sequences_v00.pdf} % default 0.92
Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

Underfull \vbox (badness 10000) has occurred while \output is active []
[18]

LaTeX Warning: File
`figures/sim_result/complex_model/complex_model7_render_comp2.pdf' not found on input line 790.

! Package pdftex.def Error: File
`figures/sim_result/complex_model/complex_model7_render_comp2.pdf' not found: using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.

...
1.790 ...ex_model/complex_model7_render_comp2.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

LaTeX Warning: File
`figures/sim_result/complex_model/complex_model9_render_com
p2.pdf' not found on input line 798.

! Package pdftex.def Error: File
`figures/sim_result/complex_model/complex_mode
l9_render_comp2.pdf' not found: using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

1.798 ...ex_model/complex_model9_render_comp2.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

LaTeX Warning: File
`figures/sim_result/complex_model/complex_model5_render_com
p2.pdf' not found on input line 806.

! Package pdftex.def Error: File
`figures/sim_result/complex_model/complex_mode
l5_render_comp2.pdf' not found: using draft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.
...

1.806 ...ex_model/complex_model5_render_comp2.pdf}

Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

LaTeX Warning: Float too large for page by 458.58304pt on input line 814.

LaTeX Warning: File
`figures/sim_result/complex_model/objectD_heatmap_comp_v3.p
df' not found on input line 825.

! Package pdftex.def Error: File
`figures/sim_result/complex_model/objectD_heat
map_comp_v3.pdf' not found: using draft setting.

1
2
3
4
5 See the pdftex.def package documentation for explanation.
6 Type H <return> for immediate help.
7 ...
8

9
10 1.825 ...omplex_model/objectD_heatmap_comp_v3.pdf}

11 Try typing <return> to proceed.
12 If that doesn't work, type X <return> to quit.
13
14

15
16 LaTeX Warning: File
17 `figures/sim_result/complex_model/objectE_heatmap_comp_v3.p
18 df' not found on input line 833.
19

20
21 ! Package pdftex.def Error: File
22 `figures/sim_result/complex_model/objectE_heat
23 map_comp_v3.pdf' not found: using draft setting.
24

25 See the pdftex.def package documentation for explanation.
26 Type H <return> for immediate help.
27 ...
28

29
30 1.833 ...omplex_model/objectE_heatmap_comp_v3.pdf}

31 Try typing <return> to proceed.
32 If that doesn't work, type X <return> to quit.
33
34

35
36 LaTeX Warning: File
37 `figures/sim_result/complex_model/objectF_heatmap_comp_v3.p
38 df' not found on input line 841.
39

40
41 ! Package pdftex.def Error: File
42 `figures/sim_result/complex_model/objectF_heat
43 map_comp_v3.pdf' not found: using draft setting.
44

45 See the pdftex.def package documentation for explanation.
46 Type H <return> for immediate help.
47 ...
48

49
50 1.841 ...omplex_model/objectF_heatmap_comp_v3.pdf}

51 Try typing <return> to proceed.
52 If that doesn't work, type X <return> to quit.
53
54

55
56 LaTeX Warning: Float too large for page by 251.13611pt on input line 850.
57
58
59
60
61
62
63
64
65

Underfull \vbox (badness 10000) has occurred while \output is active []
[19]

Underfull \vbox (badness 10000) has occurred while \output is active []
[20]

Underfull \vbox (badness 10000) has occurred while \output is active []
[21]

Underfull \vbox (badness 10000) has occurred while \output is active []
[22]

Package natbib Warning: Citation `Kaiming_2017_maskrcnn' on page 23
undefined on
input line 875.

Package natbib Warning: Citation `Kirillov_2023_ICCV' on page 23
undefined on i
nput line 875.

Package natbib Warning: Citation `Ravi_2025_sam2' on page 23 undefined on
input
line 875.

Package natbib Warning: Citation `carion_2026_sam3' on page 23 undefined
on inp
ut line 875.

Package natbib Warning: Citation `10214100' on page 23 undefined on input
line
881.

Package natbib Warning: Citation `Yang_2021_waterjet' on page 23
undefined on i
nput line 885.

Package natbib Warning: Citation `Deng_2022_DWS' on page 23 undefined on
input
line 886.

Underfull \vbox (badness 10000) has occurred while \output is active []
[23]

Package hyperref Warning: Difference (4) between bookmark levels is
greater
(hyperref) than one, level fixed on input line 904.

Package hyperref Warning: Difference (3) between bookmark levels is
greater
(hyperref) than one, level fixed on input line 907.

Package hyperref Warning: Difference (2) between bookmark levels is
greater
(hyperref) than one, level fixed on input line 910.

No file root.bbl.

Package natbib Warning: There were undefined citations.

Underfull \vbox (badness 10000) has occurred while \output is active []

[24] (./root.aux)

LaTeX2e <2024-06-01> patch level 2
L3 programming layer <2024-05-27>

LaTeX Font Warning: Size substitutions with differences
(Font) up to 0.47939pt have occurred.

Package rerunfilecheck Info: File `root.out' has not changed.
(rerunfilecheck) Checksum:
426D329510CB72467235DB3BECFD184D;8308.
)

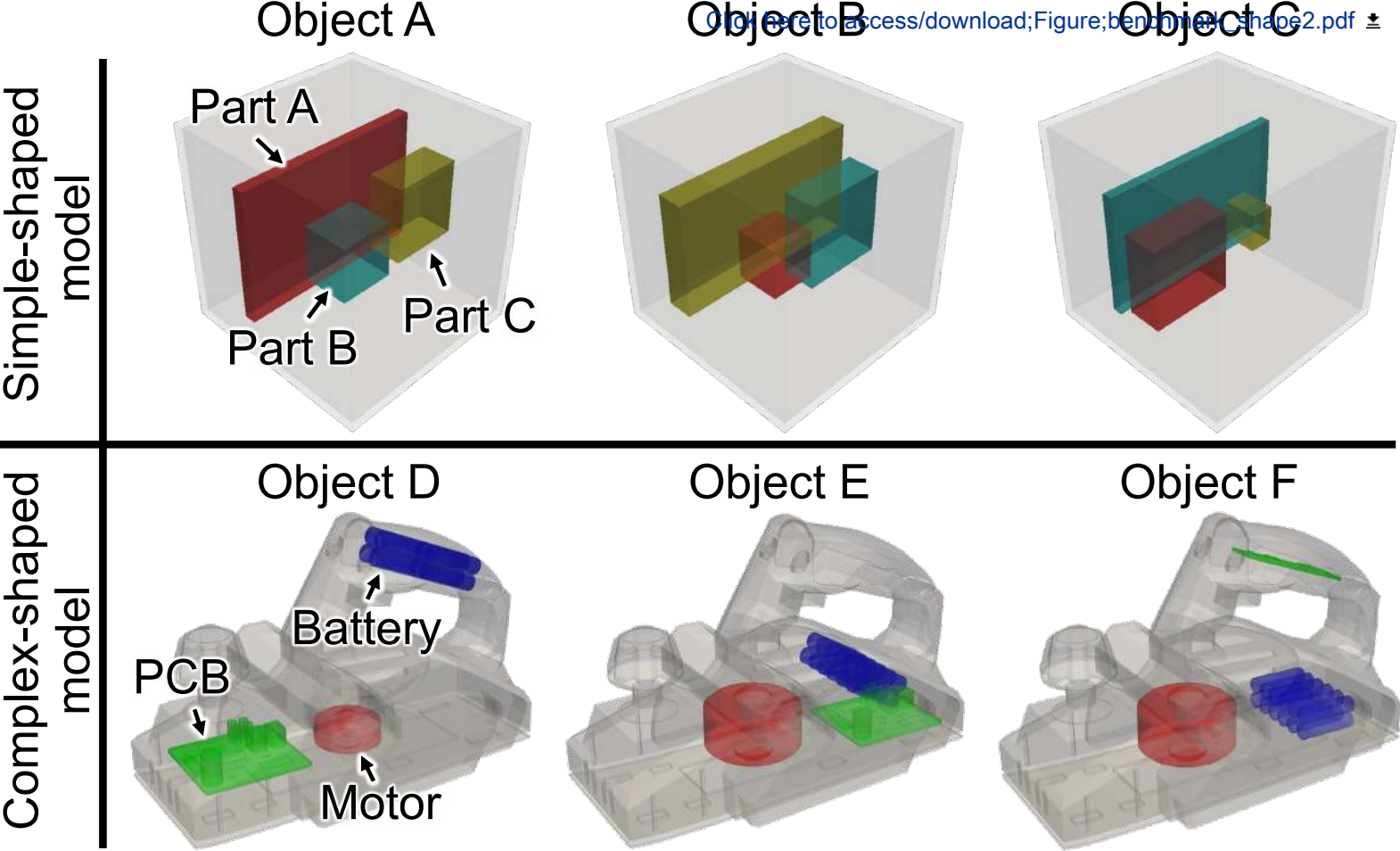
Here is how much of TeX's memory you used:
18168 strings out of 473583
314200 string characters out of 5732343
1947908 words of memory out of 5000000
40671 multiletter control sequences out of 15000+600000
582377 words of font info for 131 fonts, out of 8000000 for 9000
1151 hyphenation exceptions out of 8191
90i,12n,93p,1349b,614s stack positions out of
10000i,1000n,20000p,200000b,200000s
<c:/texlive/2024/texmf-
dist/fonts/typel/public/amsfonts/cm/cmbx10.pfb><c:/tex

```
live/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmbx12.pfb><c:/texlive/2024
/texmf-
dist/fonts/type1/public/amsfonts/cm/cmbx7.pfb><c:/texlive/2024/texmf-dis
t/fonts/type1/public/amsfonts/cm/cmbx8.pfb><c:/texlive/2024/texmf-
dist/fonts/ty
pel/public/amsfonts/cm/cmbx9.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public
/amsfonts/cm/cmex10.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts
/cm/cmmi10.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmmi5
.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmmi7.pfb><c:/t
exlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmmib10.pfb><c:/texlive/2
024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmrl10.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmrl12.pfb><c:/texlive/2024/texmf-
dist/fonts
/type1/public/amsfonts/cm/cmrl17.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/pub
lic/amsfonts/cm/cmrl5.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfont
s/cm/cmrl7.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmrl8.p
fb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmrl9.pfb><c:/texl
ive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmsy10.pfb><c:/texlive/2024/
texmf-
dist/fonts/type1/public/amsfonts/cm/cmsy7.pfb><c:/texlive/2024/texmf-dist
/fonts/type1/public/amsfonts/cm/cmti10.pfb><c:/texlive/2024/texmf-
dist/fonts/ty
pel/public/amsfonts/cm/cmtt10.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/publi
c/doublestroke/dsrom10.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfo
nts/symbols/msbm10.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/cm-
super/
sfrm0700.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/cm-
super/sfrm1000.p
fb>
```

Output written on root.pdf (24 pages, 354604 bytes).

PDF statistics:

```
475 PDF objects out of 1000 (max. 8388607)
395 compressed objects within 4 object streams
93 named destinations out of 1000 (max. 500000)
297 words of extra memory for PDF output out of 10000 (max. 10000000)
```



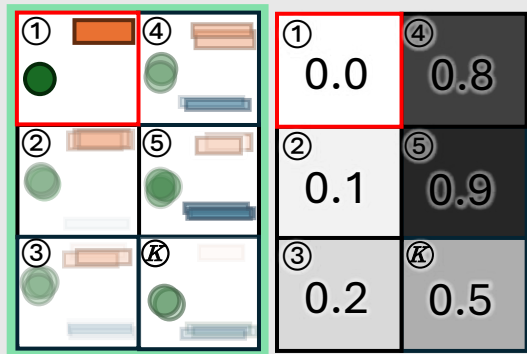
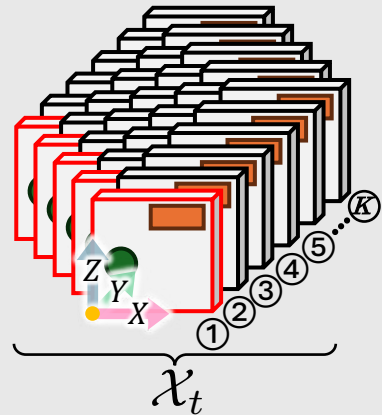
Sample
internal structure

Compute target part presence

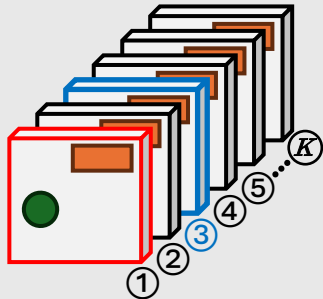
$$\mathbf{c}_t^{\text{map}} = c_{\text{ucb}}(\mathcal{X}_t)$$

Select
cutting position

$$\begin{aligned} \mathcal{A}_{\text{feasible}} &\rightarrow \{2, 3\} \\ f_{\text{vol}} &\rightarrow 3 \end{aligned} \quad \Bigg\} \rightarrow 3$$

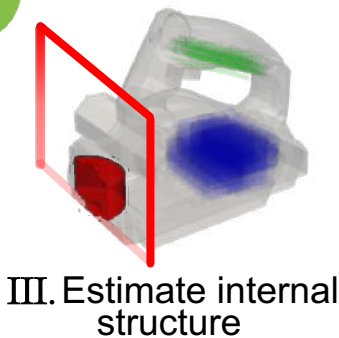
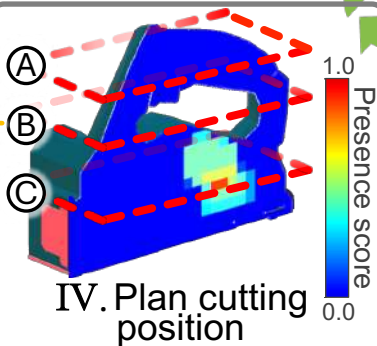
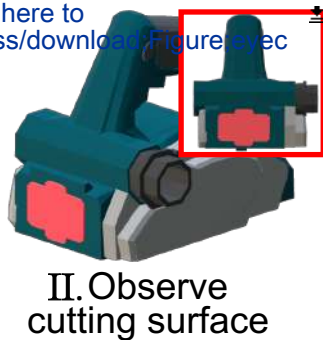
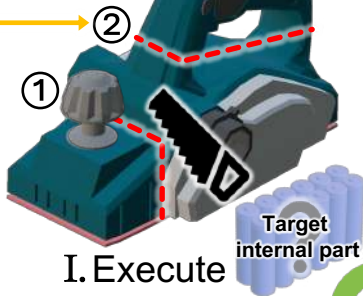


Low  High



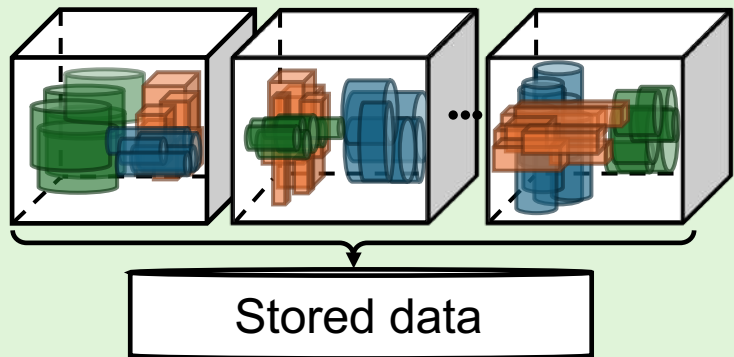
 Condition
 Ensembled internal structure
 Planned cutting position ($\mathbf{a}_{t+1}$)

[Click here to access/download/Figure/eyec](#)

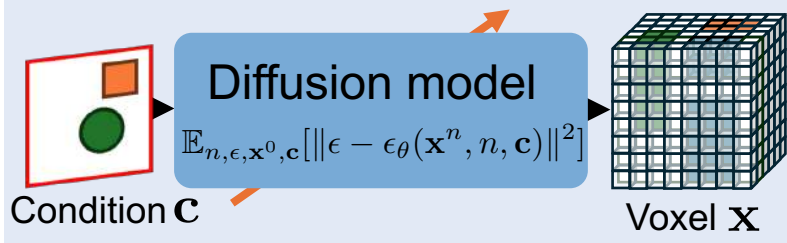


(A) Conservative (B) Optimal (C) Overcutting

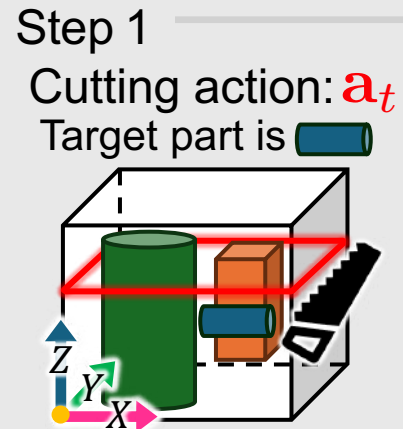
(1) Data collection: various structural patterns



(2) Model training: conditioned on cutting surfaces



(3) Deployment



Step 2

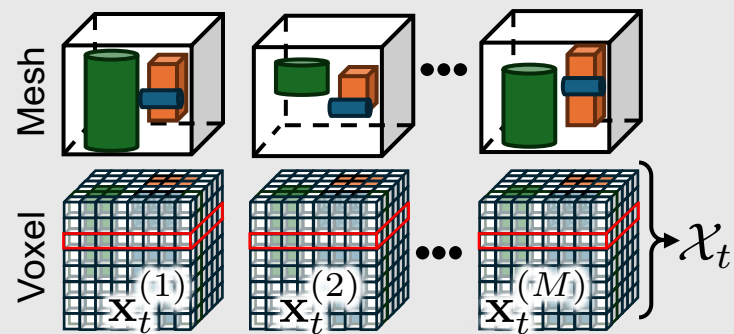
Sample M internal structures

Observed cutting surface: $\mathbf{O}_t$

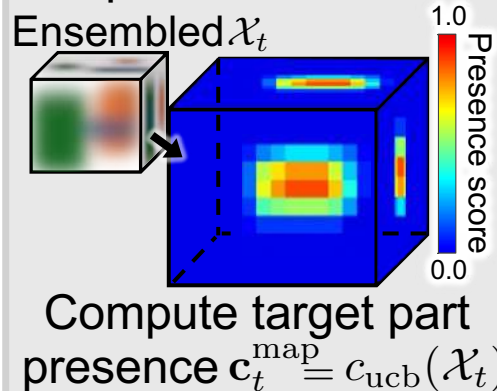
Diffusion model

$$\mathbf{x}_t^{n-1} \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t^n, n, \mathbf{c}_t), \Sigma^n)$$

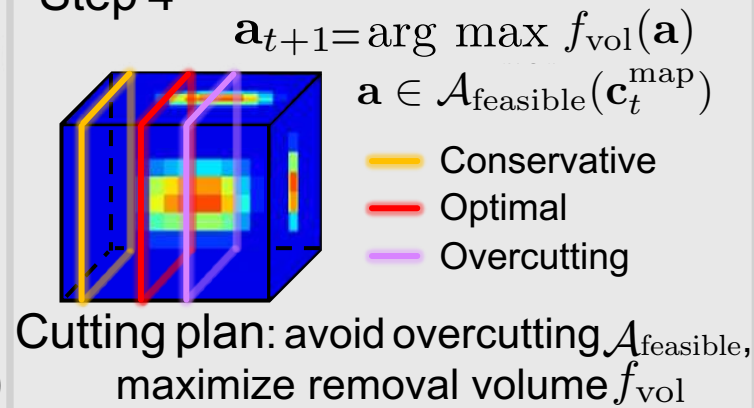
$$\text{Condition } \mathbf{c}_t \leftarrow \{\mathbf{o}_{1:t}\}$$



Step 3



Step 4



Step 5

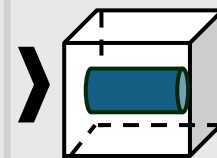
Repeat

Step 1

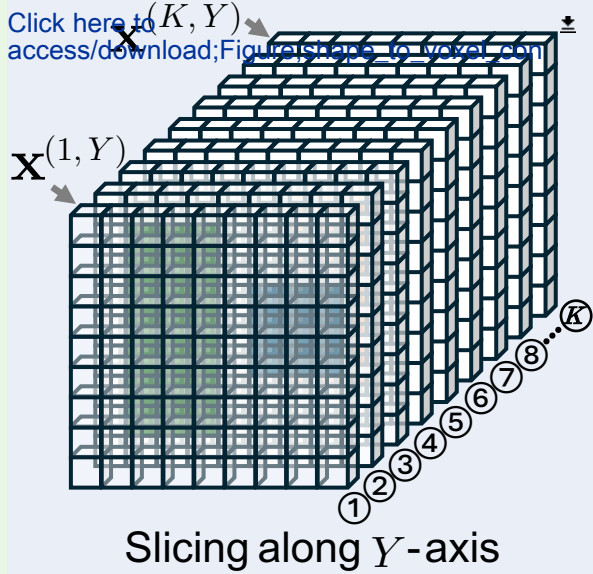
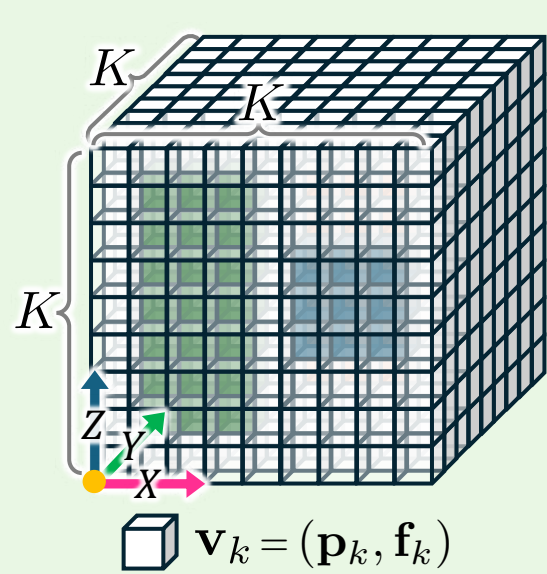
$\vdots$

Step 4

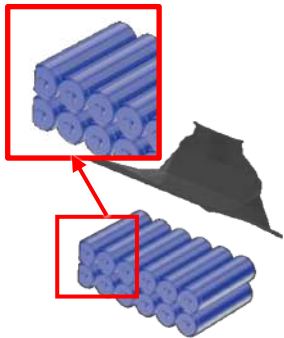
for T steps



Extract target part



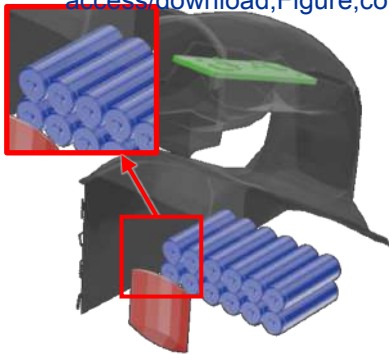
VAEAC



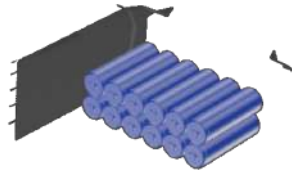
PCD-DM



Proposed

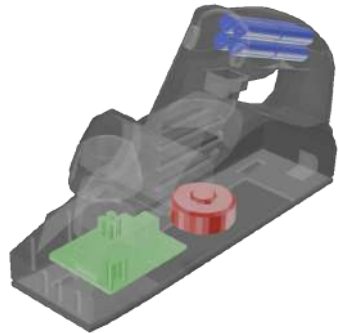


Proposed-GT_↓

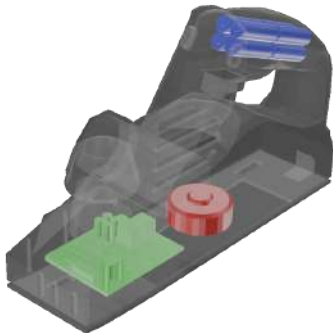


[Click here to access/download;Figure;complex_model5_render_comp2](#)

VAEAC

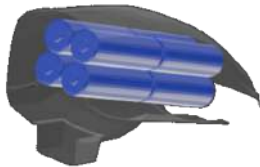


PCD-DM

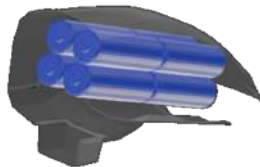


Proposed

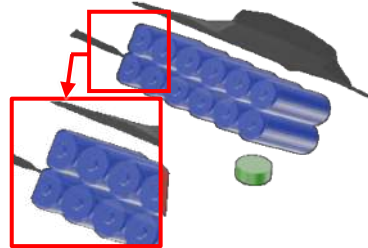
[Click here to access/download;Figure;complex_model7_render_comp2](#)



Proposed-GT_↓



VAEAC

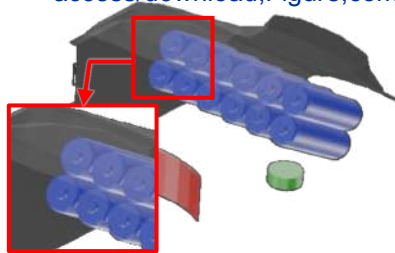


PCD-DM

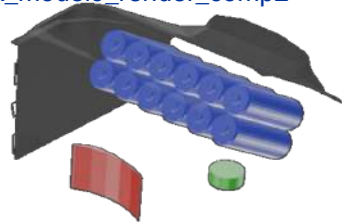


Proposed

[Click here to access/download;Figure;complex_model9_render_comp2](#)



Proposed-GT_±



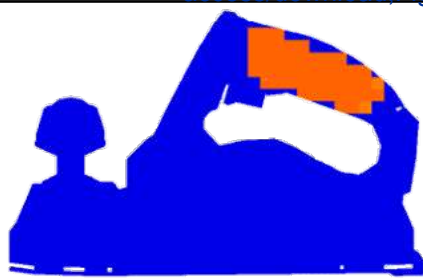
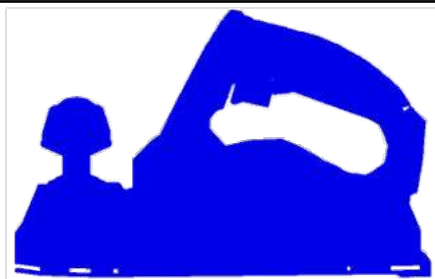


VAEAC

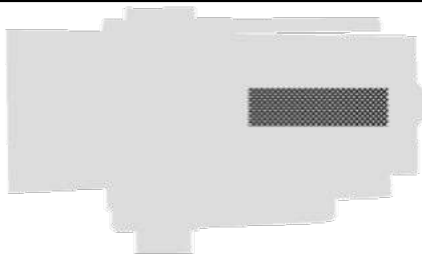
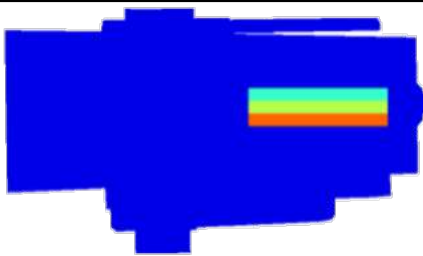
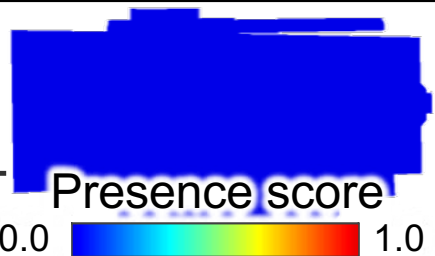
Proposed

Ground Truth 

Side view



Top view



Presence score

0.0

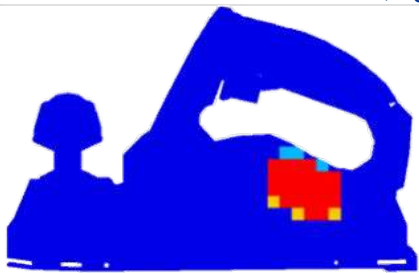
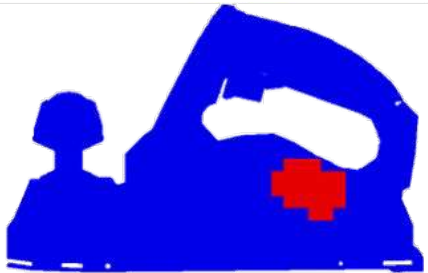
1.0

VAEAC

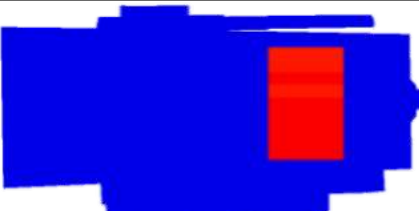
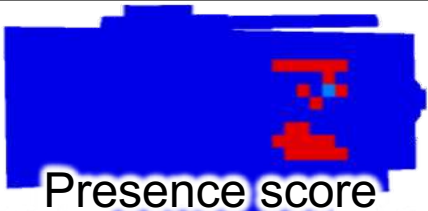
Proposed

Ground Truth 

Side view



Top view



Presence score

0.0



1.0

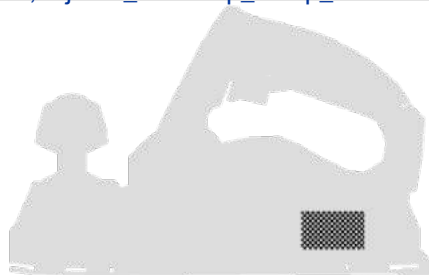
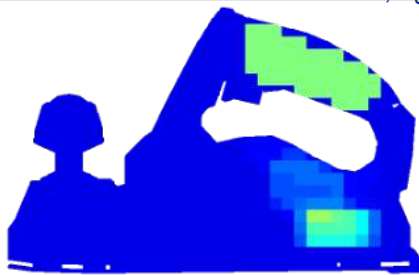
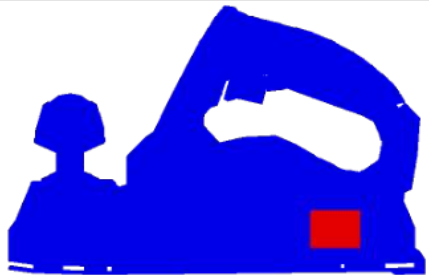
[Click here to access/download;Figure;ObjectE_heatmap_comp_v3.](#)

VAEAC

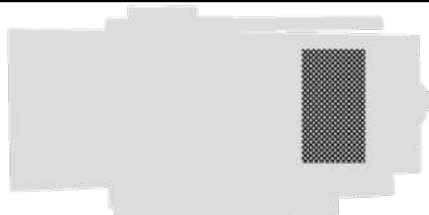
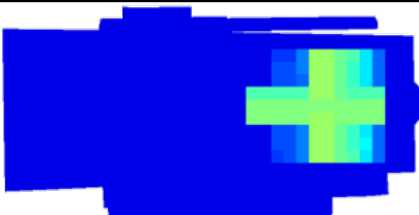
Proposed

Ground Truth $\pm$

Side view



Top view



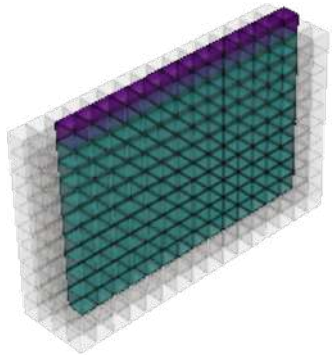
Presence score

0.0

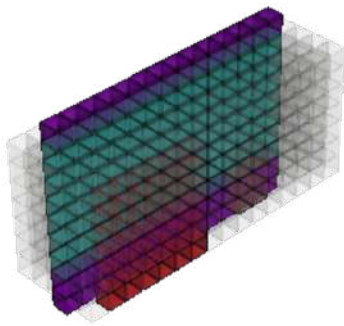


1.0

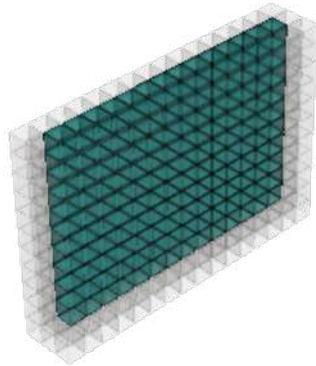
VAEAC



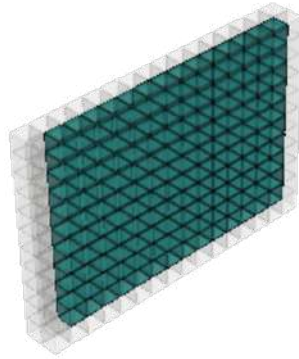
PCD-DM



Proposed

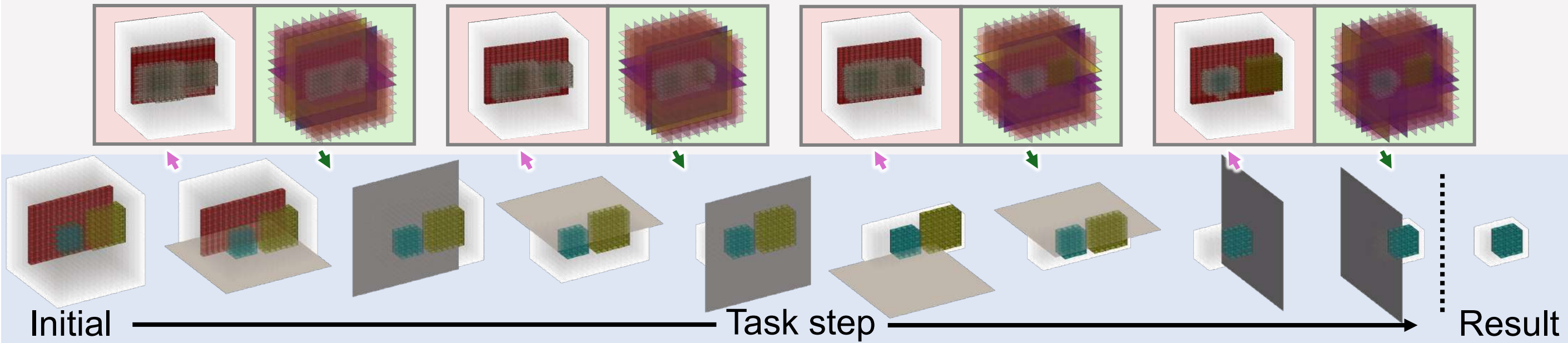


Proposed-GT



[Click here to access/download;Figure;simple_model1_render_comp_2.pdf](#) 

Estimated internal structure Cutting position candidates colored by removability (Low High) [Click here to access/download/Figure/simple_model6_cutting_process2.pdf](#)





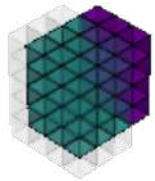
Volume of target part



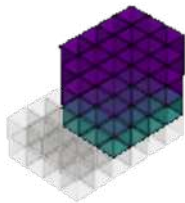
[Click here to access/download;Figure:simple_model6_render_comp_2.pdf](#) 

Overcut volume

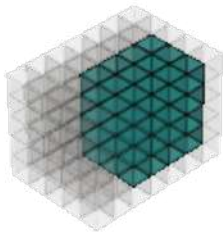
VAEAC



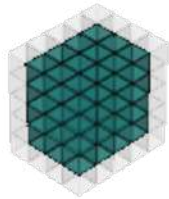
PCD-DM



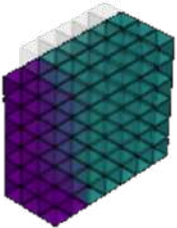
Proposed



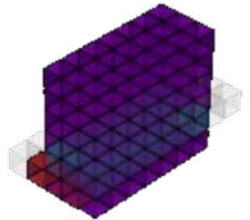
Proposed-GT



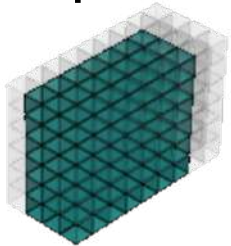
VAEAC



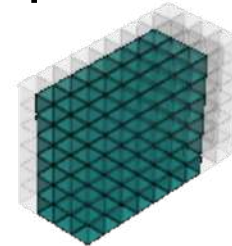
PCD-DM



Proposed



Proposed-GT



[Click here to access/download;Figure;simple_model8_render_comp_2.pdf](#) 

